



# **Nokia Service Router Linux**

## **Interface Configuration Guide Release 21.3**

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# 1 Getting started

This chapter describes this document, includes summaries of changes from previous releases and precautionary messages, and lists command conventions.

## 1.1 About this document

This document describes the interfaces used with the Nokia Service Router Linux (SR Linux). These include:

- CLI
- gNMI
- JSON

This document also provides an overview to CLI plug-ins and details Nokia defined general and operation commands, and show commands.

This document is intended for network technicians, administrators, operators, service providers, and others who need to understand how to interface with the SR Linux.



**Note:** This manual covers the current release and may also contain some content that will be released in later maintenance loads. Refer to the *SR Linux Release Notes* for information on features supported in each load.

## 1.2 What's new

This is a new document that consolidates user interface reference material and procedures. In addition, the following release-level changes were made.

**Table 1** Change summary

| Topic                       | Location                                                                                                                                          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Filtering of display output | <ul style="list-style-type: none"><li>• <a href="#">Using output filters</a></li><li>• <a href="#">General and operational commands</a></li></ul> |

**Table 1** Change summary (Continued)

| Topic                                                 | Location                                                                                                                                               |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enhanced <b>diff</b> functionality                    | <ul style="list-style-type: none"> <li>• <a href="#">Using the diff command</a></li> <li>• <a href="#">General and operational commands</a></li> </ul> |
| Environment variable for CLI default network-instance | <a href="#">Configuring the network-instance environment variable</a>                                                                                  |
| <b>copy</b> command                                   | <a href="#">Using the copy command</a>                                                                                                                 |
| <b>replace</b> command                                | <a href="#">Using the replace command</a>                                                                                                              |
| <b>watch</b> command                                  | <a href="#">Setting commands to execute periodically</a>                                                                                               |
| gNMI CAPABILITIES                                     | <a href="#">gNMI capabilities RPC</a>                                                                                                                  |
| gNMI subscribe STREAM TARGET                          | <a href="#">SubscriptionList message</a>                                                                                                               |
| Updated show reports                                  | <a href="#">Pre-defined show reports</a>                                                                                                               |

## 1.3 Precautionary messages

Observe all dangers, warnings, and cautions in this document to avoid injury or equipment damage during installation and maintenance. Follow the safety procedures and guidelines when working with and near electrical equipment.

[Table 2](#) describes information symbols contained in this document.

**Table 2** Information symbols

| Symbol                                                                              | Meaning | Description                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Danger  | Warns that incorrect handling and installation could result in bodily injury. An electric shock hazard could exist. Before you begin work on this equipment, be aware of hazards involving electrical circuitry, be familiar with networking environments, and implement accident prevention procedures. |
|  | Warning | Warns that incorrect handling and installation could result in equipment damage or loss of data.                                                                                                                                                                                                         |
|  | Caution | Warns that incorrect handling may reduce your component or system performance.                                                                                                                                                                                                                           |

**Table 2** Information symbols (Continued)

| Symbol                                                                            | Meaning | Description                                                      |
|-----------------------------------------------------------------------------------|---------|------------------------------------------------------------------|
|  | Note    | Notes contain suggestions or additional operational information. |

## 1.4 Conventions

Nokia SR Linux documentation uses the following command conventions.

- **Bold** type indicates a command that the user must enter.
- Input and output examples are displayed in `Courier` text.
- An open right-angle bracket indicates a progression of menu choices or simple command sequence (often selected from a user interface). Example: **start > connect to**.
- Angle brackets (< >) indicate an item that is not used verbatim. For example, for the command **show ethernet <name>**, *name* should be replaced with the name of the interface.
- A vertical bar (|) indicates a mutually exclusive argument.
- Square brackets ([ ]) indicate optional elements.
- Braces ({ }) indicate a required choice. When braces are contained within square brackets, they indicate a required choice within an optional element.
- *Italic* type indicates a variable.

Generic IP addresses are used in examples. Replace these with the appropriate IP addresses used in the system.

## 2 CLI interface

The CLI is an interface for configuring, monitoring, and maintaining the SR Linux. This chapter describes basic features of the CLI and how to use them.

### 2.1 Accessing and using the CLI

#### 2.1.1 Accessing the CLI

After the SR Linux device is initialized, you can access the CLI using a console or SSH connection. See the SR Linux hardware documentation for information about establishing a console connection and enabling and connecting to an SSH server.

Use the following command to connect to the SR Linux and open the CLI using SSH:

```
ssh admin@<IP Address>
```

**Example:**

```
$ ssh admin@172.16.0.3
Hello admin,
Welcome to the srlinux CLI.
Type 'help' (and press <ENTER>) if you need any help using this.
--{ running }--[ ]--
```

#### 2.1.2 Using the CLI help functions

The CLI help functions (**?** and **help**) can assist in understanding command usage and indicate which configuration mode you are in.

Enter a question mark (?) after a command to display the command usage. For example, entering a question mark after the command **network-instance**, shows its usage.

**Example 1:**

```
# network-instance ?
usage: network-instance <name>
Network instances configured on the local system
Positional arguments:
  name                [string] A unique name identifying the network instance
```

```
--{ running }--[ ]--
```

### Example 2:

Enter **help** at the top level to show the current configuration mode and details on other configuration modes. For more information about configurations, see [Configuration modes](#).

```
# help
-----
You are now in the running mode.
Here you can navigate and query the running configuration.
This configuration has been validated, committed and send to the applications.
There are multiple modes you can enter while using the CLI.
Each mode offers its own set of capabilities:
  - running mode: Allows traversing and inspecting of the running configuration.
  - state mode: Allows traversing and inspecting of the state.
  - candidate mode: Allows editing and inspecting of the configuration.
  - show mode: Allows traversing and executing of custom show routines.
To switch mode use 'enter <mode-name>', e.g. 'enter candidate'
To navigate around, you can simply type the node name to enter a context, while
'exit [all]' is used to navigate up.
'{' and '}' are an alternative way to enter and leave contexts.
'?' can be used to see all possible commands, or alternatively,
'<TAB>' can be used to trigger auto-completion.
'tree' displays the tree of possible nodes you can enter.
-----
--{ * running }--[ ]--
```

## 2.1.3 Using the CLI auto-complete function

To reduce keystrokes or aid in remembering a command name, use the CLI auto-complete function.

Enter a tab at any mode or level to auto-complete the next command level. If multiple options are available, a popup will appear.

- When a command is partially entered, the remainder of the command appears ahead of the prompt in lighter text. Press the Tab key to complete the command.

```
#info network-instance
```

- When the Tab key is pressed and there are multiple options, the options are shown:

```
# info
..      depth      from      system
/       detail     interface use-proto-json
debug   flat       network-instance
```

## 2.1.4 Using the CLI

Use shortcuts to move the cursor on the command line, complete commands, and recall commands previously entered. Shortcuts can also make syntax correction easier. [Table 3](#) lists common shortcuts.

**Table 3 CLI keyboard shortcuts**

| Task                                                                                                                                                                            | Keystroke                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Move cursor to the beginning of the line                                                                                                                                        | Ctrl+A                                              |
| Move cursor to the end of the line                                                                                                                                              | Ctrl+E                                              |
| Move cursor one character to the right                                                                                                                                          | Ctrl+F or Right arrow key                           |
| Move cursor one character to the left                                                                                                                                           | Ctrl+B or Left arrow key                            |
| Move cursor forward one word                                                                                                                                                    | Esc F                                               |
| Move cursor back one word                                                                                                                                                       | Esc B                                               |
| Transpose the character to the left of the cursor with the character the cursor is placed on                                                                                    | Ctrl+T                                              |
| Complete a partial command                                                                                                                                                      | Enter the first few letters, then press the Tab key |
| Recall previous entry in the buffer                                                                                                                                             | Page up key                                         |
| Navigate one level up within a context. For example:<br>--{running}-- [interface ethernet-1/1<br>subinterface 1]--<br># <b>exit</b><br>--{running}-- [interface ethernet-1/1]-- | Type: <b>exit</b>                                   |
| Return to the root context. For example:<br>--{running}-- [interface ethernet-1/1<br>subinterface 1]--<br># <b>exit all</b><br>--{running}-- [ ]--                              | Type: <b>exit all</b>                               |

## 2.1.5 Closing the CLI

Close the CLI using one of the following methods:

- Press Ctrl+D.

- Enter the **quit** command at the CLI prompt.

## 2.2 Configuration modes

Configuration modes define how the system is running when transactions are performed. Supported modes are the following:

- **Candidate** – Use this mode to modify a configuration. Modifications are not applied to the running system until a **commit** command is issued. When committed, the changes are copied to the running configuration and become active.

There are different types of configuration candidates. See [Configuration candidates](#).

- **Running** – Use this mode to display the currently running or active configuration. Configurations cannot be edited in this mode.
- **State** – Use this mode to display the configuration and operational states. The state mode displays more detailed information than show mode.
- **Show** – Use this mode to display configured features and operational states. The show mode displays less information than state mode.

### 2.2.1 Configuration candidates

You can modify the candidate configuration in different modes:

- Exclusive
- Shared
- Private
- Name

#### 2.2.1.1 Exclusive mode

When entering candidate mode, if you specify the **exclusive** keyword, it locks out other users from making changes to the candidate configuration.

You can enter candidate exclusive mode only under the following conditions:

- The current shared candidate configuration has not been modified.

- There are no other users in candidate shared mode.
- No other users have entered candidate exclusive mode.

### 2.2.1.2 Shared mode

By default, the candidate configuration is in shared mode. This allows multiple users to modify the candidate configuration concurrently. When the configuration is committed, the changes from all of the users are applied.

A default candidate is defined per user (see [Name mode](#)).

Use caution when allowing multiple users access to the candidate configuration at the same time. If one user commits the configuration, that commits the changes made by any other users who have modified the current candidate configuration.

### 2.2.1.3 Private mode

A private candidate allows multiple users to modify a configuration; however when a user commits their changes, only the changes from that user are committed. When a private candidate is created, private datastores are created and a snapshot is taken from the running database to create a baseline.

When starting a private candidate, a default candidate is defined per user with the name 'private-`<username>`' unless a unique name is defined (see [Name mode](#)).

### 2.2.1.4 Name mode

Candidate types support an optional name. This allows multiple users to share environments.

When a candidate is created with a name specified, a new entry is created in the /system/configuration/candidate[] list. Other users can enter the same candidate (if the candidate type is shared) using this name.

if no name is specified, then the name 'default' is used. This means the global shared candidate can be accessed by entering **enter candidate name default** or just **enter candidate**. For private candidates the name 'default' is not used (since private candidates share a list with shared candidates). Instead, private candidates are created with the name "private-`<username>`".

Named candidates are automatically deleted when there are no active sessions present and they are empty, or after 7 days of no session activity. This 7-day default is configurable in the `/system/configuration/idle-timeout` field.

## 2.2.2 Setting the configuration mode

After logging in to the CLI, you are initially placed in running mode. [Table 4](#) describes the commands to change between modes.

**Table 4** Commands to change configuration mode

| To enter this mode:                        | Type this command:                                 |
|--------------------------------------------|----------------------------------------------------|
| Candidate shared                           | <b>enter candidate</b>                             |
| Candidate mode for named shared candidate  | <b>enter candidate name &lt;name&gt;</b>           |
| Candidate private                          | <b>enter candidate private</b>                     |
| Candidate mode for named private candidate | <b>enter candidate private name &lt;name&gt;</b>   |
| Candidate exclusive                        | <b>enter candidate exclusive</b>                   |
| Exclusive mode for named candidate         | <b>enter candidate exclusive name &lt;name&gt;</b> |
| Running                                    | <b>enter running</b>                               |
| State                                      | <b>enter state</b>                                 |
| Show                                       | <b>enter show</b>                                  |

### Example 1:

To change from running to a shared candidate mode (using the default)':

```
--{ running }--[ ]--  
# enter candidate  
--{ * candidate shared default }--[ ]--
```

The asterisk (\*) next to the mode name indicates that the candidate configuration has changes that have not yet been committed.

### Example 2:

To switch between candidate shared and candidate exclusive modes, you must first switch to a different configuration mode (for example, running mode) before entering candidate shared or exclusive mode. For example:

```
--{ running }--[ ]--
# enter candidate exclusive
--{ candidate exclusive }--[ ]--
Are you sure? (y/[n]):
# enter running
--{ running }--[ ]--
# enter candidate
--{ candidate shared default }--[ ]--
```

### Example 3:

To enter candidate mode for a named configuration candidate, you specify the name of the configuration candidate. For example:

```
--{ running }--[ ]--
# enter candidate name cand1
--{ candidate shared cand1 }--[ ]--
```

## 2.2.3 Managing configuration conflicts

When a user enters candidate mode, the system creates two copies of the running datastore: one is modifiable by the user, and the other serves as a baseline. The modifiable datastore and the baseline datastore are collectively known as a configuration candidate.

The **baseline** command can assist in managing conflicts and uses the following arguments:

- **baseline update** - Performs an update of the complete baseline datastore, pulling in any changes that occurred in the running datastore since the baseline snapshot was taken.
- **baseline diff** - Shows baseline configuration changes with options to refine by area.
- **baseline check** - Performs a dry-run baseline check, and if conflicts are detected, an informational or warning message is generated.

## 2.2.4 Committing a configuration in candidate mode

Changes made during a configuration modification session do not take effect until a **commit** command is issued. Use the **commit** command in candidate mode only.

1. Enter candidate mode:  
    # **enter candidate**
2. Enter configuration commands.
3. Enter the **commit** command when with the required option (see [Table 5](#)).

**Table 5**      **commit command options**

| Option                                                                                                    | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Permitted additional arguments                        |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>commit now</b>                                                                                         | Apply the changes, exit candidate mode, and enter running mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | NA                                                    |
| <b>commit stay</b>                                                                                        | Apply the changes and then remain in candidate mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | commit stay [save] [comment] [confirmed]              |
| <b>commit save</b>                                                                                        | Apply the changes and automatically save the commit to the startup configuration. Can be used other arguments except <b>now</b> (e.g., <b>commit stay save</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                     | commit [stay] [checkpoint] save [confirmed] [comment] |
| <b>commit checkpoint</b>                                                                                  | Apply the changes and cause an automatic checkpoint after the commit succeeds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | commit [stay] [now] checkpoint [save] [confirmed]     |
| <b>commit validate</b>                                                                                    | Verify that a propose configuration change passes a management server validation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | NA                                                    |
| <b>commit comment &lt;comment&gt;</b>                                                                     | Use with other keywords (except <b>validate</b> ) to add a user comment (e.g. <b>commit stay comment &lt;comment&gt;</b> where <b>&lt;comment&gt;</b> is a quoted string, 1-255 characters).                                                                                                                                                                                                                                                                                                                                                                                          | commit [stay] [save] [checkpoint] [confirmed] comment |
| <b>commit confirmed</b><br><b>commit confirmed [timeout]</b><br><b>commit confirmed [accept   reject]</b> | <p>Apply the changes, but requires an explicit confirmation to become permanent. If the explicit confirmation is not issued within a specified time period, all changes are automatically reverted.</p> <p>The timeout period default is 600 seconds (10 mins.), or can be provisioned with a value of 1-86400 sec.). The timeout parameter cannot be used with the accept or reject parameter.</p> <p>Prior to the timer expiring, the accept parameter explicitly confirms and applies the changes. With no timer running, the reject parameter explicitly rejects the changes.</p> | commit [checkpoint] [save] [stay] [comment] confirmed |

### Example 1:

This example shows the **commit stay** option:

```
# enter candidate
--{ candidate shared default}--[ ]--
# interface ethernet-1/1 subinterface 1
--{ * candidate shared default}--[ interface ethernet-1/1 subinterface 1 ]--

# commit stay
All changes have been committed. Starting new transaction.

--{ candidate shared default}--[ interface ethernet-1/1 subinterface 1 ]--
```

### Example 2:

This example shows the **commit confirmed** option with a custom timeout followed by an accept action.

```
--{ * candidate shared default}--[ ]--
# commit confirmed timeout 86400
Commit confirmed (automatic rollback in a day)
All changes have been committed. Leaving candidate mode.
--{ running }--[ ]--

# commit confirmed accept
Info: Commit confirmed, automatic rollback cancelled
--{ candidate shared default}--[ ]--
#
```

## 2.2.5 Deleting configurations

Use the **delete** command to delete configurations while in candidate mode.

### Example:

The following example displays the system banner configuration, deletes the configured banner, then displays the resulting system banner configuration:

```
--{ candidate shared default}--[ ]--
# info system banner
  system {
    banner {
      login-banner "Welcome to SRLinux!"
    }
  }
--{ candidate shared default}--[ ]--
# delete system banner
--{ candidate shared default}--[ ]--
# info system banner
  system {
```

```

        banner {
        }
    }

```

## 2.2.6 Annotating the configuration

To aid in reading a configuration, you can add comments or descriptive annotations. The annotations are indicated by **!!!** in displayed output.

You can enter a comment either directly from the command line or by navigating to a CLI context and entering the comment in annotate mode.

### Example 1:

The following example adds a comment to an ACL configuration. If there is already a comment in the configuration, the new comment is appended to the existing comment.

```

--{ candidate shared default}--[ ]--
# acl ipv4-filter ip_tcp !! "Filter TCP traffic"

```

To replace the existing comment, use **!!!** instead of **!!** in the command.

### Example 2:

The following example adds the same comment to the ACL by navigating to the context for the ACL and entering the comment in annotate mode:

```

--{ * candidate shared default}--[ ]--
# acl ipv4-filter ip_tcp
--{ * candidate shared default}--[ acl ipv4-filter ip_tcp ]--
# annotate
Press [Meta+enter] or [Esc] followed by [Enter] to finish
-> Filter TCP traffic

```

You can enter multiple lines in annotate mode. To exit annotate mode, press Esc, then the Enter key.

In CLI output, the comment is displayed in the context it was entered. For example:

```

--{ running }--[ ]--
# info acl
  acl {
    ipv4-filter ip_tcp {
      !!! Filter TCP traffic
      entry 100 {
        action {
          drop {
            log true
          }
        }
      }
    }
  }

```

```

    }
  }
}
entry 110 {
  action {
    accept {
      log true
    }
  }
}
}
}
}

```

To remove a comment, enter annotate mode for the context and press Esc then Enter without entering any text.

## 2.2.7 Discarding a configuration in candidate mode

You can discard previously applied configurations with the **discard** command. Use the **discard** command in candidate mode only.

- To discard the changes and remain in candidate mode with a new candidate session, enter **discard stay**.
- To discard the changes, exit candidate mode, and enter running mode, enter **discard now**.

### Example:

All changes have been committed. Starting new transaction.

```

--{ candidate shared }--[ interface ethernet-1/1 subinterface 1 ]--
# discard stay
--{ candidate shared }--[ interface ethernet-1/1 subinterface 1 ]--

```

## 2.3 Using administrative commands

The common administrative CLI commands in this section can help you understand a current configuration and perform routine configuration tasks.

### 2.3.1 Pinging a destination IP address

Use the **ping** (IPv4) or **ping6** (IPv6) command to contact an IP address. Use this command in any mode.

**Example:**

```
--{ running }--[ ]--
# ping 192.168.1.1 network-instance default
Pinging 192.168.1.1 in srbase-default
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.027 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.032 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.030 ms
^C
--- 192.168.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6165ms
rtt min/avg/max/mdev = 0.027/0.030/0.033/0.005 ms
```

### 2.3.2 Tracing the path to a destination

To display the path a packet takes to a destination, use the **traceroute** (IPv4) or **traceroute6** (IPv6) command.

**Example:**

```
--{ running }--[ ]--
# traceroute 1.1.1.1 network-instance mgmt
Using network instance srbase-mgmt
traceroute to 1.1.1.1 (1.1.1.1), 30 hops max, 60 byte packets
 1 172.18.18.1 (172.18.18.1) 1.268 ms 1.260 ms 1.256 ms
 2 172.21.40.1 (172.21.40.1) 1.253 ms 1.848 ms 1.851 ms
 3 172.22.35.230 (172.22.35.230) 1.835 ms 1.834 ms 1.828 ms
 4 66.201.62.1 (66.201.62.1) 3.222 ms 3.222 ms 3.216 ms
 5 66.201.34.17 (66.201.34.17) 5.474 ms 5.475 ms 5.480 ms
 6 * * *
 7 206.81.81.10 (206.81.81.10) 32.577 ms 32.542 ms 32.400 ms
 8 1.1.1.1 (1.1.1.1) 22.627 ms 22.637 ms 22.638 ms
```

To trace the route using TCP SYN packets rather than UDP or ICMP echo packets, use the **tcptracert** command.

### 2.3.3 Configuring the network-instance environment variable

When you enter administrative commands such as **ping** and **traceroute**, you can specify the network-instance to be used with the command. If you do not specify a network-instance, then the network-instance configured in the network-instance environment variable is used.

If you do not specify a network-instance, or the network-instance cannot be inferred from the CLI context, then the “<base>” network instance is used when no network-instance is specified in a **ping** or **traceroute** command; for example:

```
--{ running }--[ ]--
# ping 192.168.1.1
Using network instance <base>
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.027 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.032 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.030 ms
^C
--- 192.168.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6165ms
rtt min/avg/max/mdev = 0.027/0.030/0.033/0.005 ms
```

To configure the network-instance environment variable, use the **environment network-instance** command; for example:

```
--{ candidate shared default }--[ ]--
# environment network-instance red
```

In this example, the network-instance environment variable is set to “red”, so network instance red is implied when the **ping** or **traceroute** command is entered without a network-instance name; for example:

```
--{ running }--[ ]--
# ping 192.168.1.1
Using network instance srbase-red
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.027 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.032 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.030 ms
^C
--- 192.168.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6165ms
rtt min/avg/max/mdev = 0.027/0.030/0.033/0.005 ms
```

## 2.3.4 Using bash mode

From within the CLI, use the **bash** command to enter the bash shell. Use this command in any mode. To exit the bash shell and return to the CLI, enter **exit**.

### Example:

```
--{ running }--[ ]--
# bash
[root@3-node_srlinux-A /]# ls
anaconda-
post.log  dev          etc  lib  media  opt  root  sbin  sys  tmp  var
bin       entrypoint.sh  home lib64 mnt    proc run  srv  tini usr
[root@3-node_srlinux-A /]# exit
logout
--{ running }--[ ]--
#
```

You can use the **bash** command to enter Linux commands directly from the SR Linux CLI prompt. For example:

```
--{ running }--[ ]--
# bash cat /etc/hosts
127.0.0.1    localhost
::1         localhost ip6-localhost ip6-loopback
fe00::0     ip6-localnet
ff00::0     ip6-mcastprefix
ff02::1     ip6-allnodes
ff02::2     ip6-allrouters
172.18.18.5  3-node-srlinux-A
### SRLINUX - ANYTHING MODIFIED BEYOND THIS POINT WILL BE OVERWRITTEN ###
127.0.0.1 3-node-srlinux-A
### SRLINUX FOOTER ###
```

## 2.3.5 Setting commands to execute periodically

To set a command to execute periodically, use the **watch** command. The **watch** command can be used with any valid command with the exception of interactive commands (for example, **ping**, **monitor**, **packet-trace**, **bash**, and bash oneliners such as **bash cat /etc/hosts**). An error message is generated if an interactive command is used. Output redirection is supported.

When used, the first page of output displays. The command then re-executes every 2 seconds by default.

The watch command uses the following format:

```
watch <arguments> <command to watch>
```

All arguments must precede the command being watched. Arguments for the watch command are:

| Arguments              | Definition                                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| interval               | Sets an interval for the command to re-execute.<br>Range: 1 - 3600 seconds. Default: 2 seconds                         |
| differences            | For output that is actively changing, highlights the differences between the previous and current execution            |
| cumulative-differences | For output that is actively changing, highlights the differences between the first execution and the current execution |
| no-colors              | When used with differences or cumulative-differences, markers are used to show differences (verses colors)             |
| no-limit               | Allows the full output of each execution to display (verses limiting to the height of the terminal)                    |
| no-paging              | Provides continuous output (verses redrawing the terminal with each execution)                                         |
| no-title               | Turns off the header that shows the interval, command, number of executions that have occurred, and the time           |

The command being watched does not require double quotes. In addition, the auto complete function can be used to complete a valid command.

To exit the **watch** command, enter Ctrl-C.

### Example 1:

In the following example, the watch command is used to show interface statistics (showing in-octets and out-octets in table format). The interval is modified to 10 seconds.

```
# watch interval 10 info from state interface * statistics | as table | filter fields
in-octets out-octets
```

```
Every 10.0s: info from state interface * statistics | as table | filter fields in-
octets out-octets (Executions 2,
Thu 04:47:34PM)
```

| Interface     | In-octets | Out-octets |
|---------------|-----------|------------|
| ethernet-1/1  | 812755    | 812626     |
| ethernet-1/2  | 811411    | 810362     |
| ethernet-1/3  |           |            |
| ethernet-1/4  |           |            |
| ethernet-1/5  |           |            |
| ethernet-1/6  |           |            |
| ethernet-1/7  |           |            |
| ethernet-1/8  |           |            |
| ethernet-1/9  |           |            |
| ethernet-1/10 |           |            |
| ethernet-1/11 |           |            |
| ethernet-1/12 |           |            |
| ethernet-1/13 |           |            |
| ethernet-1/14 |           |            |
| ethernet-1/15 |           |            |

### Example 2:

In the following example, the watch command is used to show differences in interface statistics (with in-octets and out-octets in table format). In addition, the no-color argument is used to show the differences between previous and current execution with markers verses color indicators.

```
# watch differences no-colors info from state interface * statistics | as table
| filter fields in-octets out-octets
```

```
Every 10.0s: info from state interface * statistics | as table | filter fields in-
octets out-octets (Executions 4,
Thu 04:54:48PM)
```

| Interface    | In-octets | Out-octets |
|--------------|-----------|------------|
| ethernet-1/1 | 817889    | 817760     |
| ethernet-1/1 | 817975    | 817865     |
| ethernet-1/2 | 816328    | 815088     |
| ethernet-1/2 | 816585    | 815278     |
| ethernet-1/3 |           |            |

|              |  |  |  |
|--------------|--|--|--|
| ethernet-1/4 |  |  |  |
| ethernet-1/5 |  |  |  |
| ethernet-1/6 |  |  |  |
| ethernet-1/7 |  |  |  |
| ethernet-1/8 |  |  |  |
| ethernet-1/9 |  |  |  |

## 2.3.6 Using the diff command

Use the **diff** command to get a comparison of configuration changes. Optional arguments can be used to indicate the source and destination datastore. The following use rules apply:

- If no arguments are specified, the diff is performed from the candidate to the baseline of the candidate.
- If a single argument is specified, the diff is performed from the current candidate to the specified candidate.
- If two arguments are specified, the first is treated as the source, and the second as the destination.

Global arguments include: baseline, candidate, checkpoint, factory, file, from, rescue, running, and startup.

The **diff** command can be used outside of candidate mode, but only if used with arguments.

### Example 1:

The following shows a basic **diff** command without arguments. In this example, the description and admin state of an interface are changed and the differences shown:

```
--{ candidate shared default }--[ ]--
# interface ethernet-1/1 admin-state disable
--{ * candidate shared default }--[ ]--
# interface ethernet-1/2 description "updated"
--{ * candidate shared default }--[ ]--
# diff
+ interface ethernet-1/1 {
+   admin-state disable
+ }
+ interface ethernet-1/2 {
+   description updated
+ }
```

## Example 2:

The following shows a diff with a single argument. In this example, the comparison occurs to and from a factory configuration.

```
--{ candidate shared default }--[ ]--
# diff factory
    acl {
-
+     ipv4-filter allow_sip_dip {
+         subinterface-specific output-only
+         entry 10 {
+             action {
+                 accept {
+                 }
+             }
+             match {
+                 destination-ip {
+                     prefix 100.1.2.2/32
+                 }
+                 source-ip {
+                     prefix 100.1.1.1/32
+                 }
+             }
+         }
+     }
+     entry 100 {
+         action {
+             accept {
+             }
+         }
+     }
+ }
```

## Example 3:

The following shows an example where the comparison occurs between two checkpoints.

```
--{ candidate shared default }--[ ]--
# save checkpoint name current
/system:
    Generated checkpoint '/etc/opt/srlinux/checkpoint/checkpoint-0.json' with name 'current' and comment ''

--{ candidate shared default }--[ ]--
# interface ethernet-1/1 description "changed-in-next-checkpoint"
--{ candidate shared default }--[ ]--
# commit stay
All changes have been committed. Starting new transaction.
--{ candidate shared default }--[ ]--
# save checkpoint name newer
/system:
    Generated checkpoint '/etc/opt/srlinux/checkpoint/checkpoint-0.json' with name 'newer' and comment ''

--{ candidate shared default }--[ ]--
# diff checkpoint newer checkpoint current
```

```
-      interface ethernet-1/1 {
+          description changed-in-next-checkpoint
+          description dut1-dut2-1
+      }
```

## 2.3.7 Using the copy command

Use the **copy** command to copy a specific context to another context. For example, you can use this command to copy a container or any configuration hierarchy and give it another name (such as an interface).

The **copy** command merges rather than replaces the existing context. If a replace is required, you can delete the target node first. When using the **copy** command, matching fields populate to the destination node; fields that do not match are ignored. The use of ranges for both source and destination is permitted.

Use this command in candidate mode. Nokia also recommends that you use the **diff** command to verify changes when using complex copy operations such as multiple ranges in both the source and destination.

### Example 1:

The following shows the context of two existing interfaces (ethernet-1/1 and ethernet-2/1):

```
--{ candidate shared default }--[ ]--
# info interface ethernet-1/1
    interface ethernet-1/1 {
        description dut1-dut2-1
        ethernet {
            aggregate-id lag1
        }
    }
--{ candidate shared default }--[ ]--
# info interface ethernet-2/1
    interface ethernet-2/1 {
        description dut2-dut4-1
        subinterface 1 {
            ipv4 {
                address 1.2.4.2/24 {
                }
            }
            ipv6 {
                address 3ffe:0:2:4::2/64 {
                }
            }
        }
    }
}
```

## Example 2:

To copy the context of ethernet-1/1 to ethernet-2/1, enter the following:

```
--{ candidate shared default }--[ ]--
# copy interface ethernet-1/1 to interface ethernet-2/1
--{ * candidate shared default }--[ ]--
# diff
    interface ethernet-2/1 {
-        description dut2-dut4-1
+        description dut1-dut2-1
        ethernet {
+            aggregate-id lag1
        }
    }
```

## Example 3:

The following shows an example where only some fields match. Fields that do not match are ignored:

```
--{ * candidate shared default }--[ ]--
# info acl cpm-filter ipv4-filter entry 10
    acl {
        cpm-filter {
            ipv4-filter {
                entry 10 {
                    description "cpm-filter description"
                    action {
                        accept {
                        }
                    }
                    match {
                        protocol icmp
                        icmp {
                            type dest-unreachable
                            code [
                                0
                                1
                                2
                                3
                                4
                                13
                            ]
                        }
                    }
                }
            }
        }
    }

--{ * candidate shared default }--[ ]--
# copy acl cpm-filter ipv4-filter entry 10 to interface ethernet-2/1
--{ * candidate shared default }--[ ]--
A:dev-dut2# diff
    interface ethernet-2/1 {
-        description dut2-dut4-1
+        description "cpm-filter description"
```

}

#### Example 4:

The following shows an example that uses ranges. The system expands both the source and destination to determine which set has the largest number of keys, and copies 1-to-1 until the larger of the two ranges finishes.

```
--{ * candidate shared default }--[ ]--
# copy interface ethernet-1/{1,2} to interface ethernet-{2,3}/{1..10}
--{ * candidate shared default }--[ ]--
# diff
    interface ethernet-2/1 {
-       description dut2-dut4-1
+       description dut1-dut2-1
+       ethernet {
+           aggregate-id lag1
+       }
    }
    interface ethernet-2/2 {
-       description dut2-dut3-1
+       description "second description"
+       mtu 9000
+   }
+   interface ethernet-2/3 {
+       description dut1-dut2-1
+       ethernet {
+           aggregate-id lag1
+       }
+   }

+   interface ethernet-3/1 {
+       description dut1-dut2-1
+       ethernet {
+           aggregate-id lag1
+       }
+   }
+   interface ethernet-3/2 {
+       description "second description"
+       mtu 9000
+   }
+   interface ethernet-3/3 {
+       description dut1-dut2-1
+       ethernet {
+           aggregate-id lag1
+       }
+   }
+   interface ethernet-3/4 {
+       description "second description"
+       mtu 9000
+   }

+   interface ethernet-3/9 {
+       description dut1-dut2-1
+       ethernet {
+           aggregate-id lag1
+       }
+   }
```

```
+    }
+    interface ethernet-3/10 {
+        description "second description"
+        mtu 9000
+    }
```

## 2.3.8 Using the replace command

Use the **replace** command to replace a source context with a destination context (including all of its children). Both the source and destination can be text. This command can also be used to create a destination that does not currently exist by creating a new key that uses the source configuration.

The **replace** command has the following usage: `replace <original text> with <replacement text>`

Use this command in candidate mode.

### Example 1:

The following example replaces all instances of the "mgmt" network-instance with a "test" network-instance:

```
--{ candidate shared default }--[ ]--
# replace "network-instance mgmt" with "network-instance test"
--{ * candidate shared default }--[ ]--
# diff
    system {
        gnmi-server {
-            network-instance mgmt {
+            network-instance test {
                admin-state enable
                port 50052
                tls-profile tls-profile-1
            }
        }
        json-rpc-server {
-            network-instance mgmt {
+            network-instance test {
                http {
                    admin-state enable
                    use-authentication false
                    port 4000
                }
                https {
                    admin-state enable
                    port 4001
                    tls-profile tls-profile-1
                }
            }
        }
        dns {
```

```
-          network-instance mgmt
+          network-instance test
```

### Example 2:

The following example shows a command line that can be used to replace all instances of "interface lag3" or "aggregate-id lag3" with "interface lag2" or "aggregate-id lag2":

```
--{ candidate shared default }--[ ]--
# replace "(interface |aggregate-id )lag3" with \1lag2
```

### Example 3:

The **replace** command can also be used with **environment alias** command so that multiple arguments can be used. In the following, an alias named 'ifrename' is created to replace the name of an interface:

```
--{ candidate shared default }--[ ]--
# environment alias ifrename "replace \"(interface |aggregate-id ){{\\b\\\" with \\1{ } /
"
```

Once the alias is created, the alias with the arguments is entered.

```
--{ candidate shared default }--[ ]--
# ifrename lag2 lag9
```

For additional information on the use of aliases, see [Configuring CLI command aliases](#).

## 2.3.9 Displaying configuration details

Use the **info** command to display the configuration. Entering the **info** command from the root context displays the entire configuration, or the configuration for a specified context. Entering the command from within a context limits the display to the configuration under that context. Use this command in candidate or running mode.

### Example 1:

To display the entire configuration, enter **info** from the root context:

```
--{ candidate shared default }--[ ]--
# info
<all the configuration is displayed>
--{ candidate }--[ ]--
```

## Example 2:

To display the configuration for a specific context, enter **info** and specify the context:

```
--{ candidate shared default }--[ ]--
# info system lldp
  system {
    lldp {
      admin-state enable
      hello-timer 600
      management-address mgmt0.0 {
        type [
          IPv4
        ]
      }
      interface mgmt0 {
        admin-state disable
      }
    }
  }
--{ candidate }--[ ]--
```

## Example 3:

From a context, use the **info** command to display the configuration under that context:

```
--{ candidate shared default }--[ ]--
# system lldp
--{ candidate }--[ system lldp ]--
# info
  admin-state enable
  hello-timer 600
  management-address mgmt0.0 {
    type [
      IPv4
    ]
  }
  interface mgmt0 {
    admin-state disable
  }
--{ candidate }--[ system lldp ]--
```

Use the **as-json** option to display JSON-formatted output:

```
--{ candidate }--[ system lldp ]--
# info | as json
{
  "admin-state": "enable",
  "hello-timer": "600",
  "management-address": [
    {
      "subinterface": "mgmt0.0",
      "type": [
        "IPv4"
      ]
    }
  ]
}
```

```
    }
  ],
  "interface": [
    {
      "name": "mgmt0",
      "admin-state": "disable"
    }
  ]
}
```

#### Example 4:

Use the **detail** option to display values for all parameters, including those not specifically configured:

```
--{ candidate shared default }--[ system lldp ]--
# info detail
  admin-state enable
  hello-timer 600
  hold-multiplier 4
  management-address mgmt0.0 {
    type [
      IPv4
    ]
  }
  interface mgmt0 {
    admin-state disable
  }
```

#### Example 5:

Use the **flat** option to display the output as a series of **set** statements, omitting indentation for any sub-contexts:

```
--{ candidate shared default }--[ system lldp ]--
# info flat
set / system lldp admin-state enable
set / system lldp hello-timer 600
set / system lldp management-address mgmt0.0
set / system lldp management-address mgmt0.0 type [ IPv4 ]
set / system lldp interface mgmt0
set / system lldp interface mgmt0 admin-state disable
```

#### Example 6:

Use the **depth** option to display parameters with a specified number of sub-context levels:

```
--{ candidate shared default }--[ system lldp ]--
# info depth 0
  admin-state enable
  hello-timer 600

--{ candidate shared default }--[ system lldp ]--
```

```
# info depth 1
  admin-state enable
  hello-timer 600
  management-address mgmt0.0 {
    type [
      IPv4
    ]
  }
  interface mgmt0 {
    admin-state disable
  }
```

## 2.3.10 Displaying the command tree hierarchy

Use the **tree** command to display the tree hierarchy for all available nodes you can enter. Entering the **tree** command from the root context displays the entire tree hierarchy. Entering the command from a context limits the display to the nodes under that context. Use this command in candidate or running mode.

```
--{ candidate shared default}--[ ]--
# tree
<root>
+-- acl
|   +-- ipv4-filter
|   |   +-- description
|   |   +-- subinterface-specific
|   |   +-- statistics-per-entry
|   |   +-- entry
|   |       +-- description
|   |       +-- action
|   |           +-- accept
|   |           |   +-- log
|   |           +-- drop
|   |               +-- log
|   +-- match
|       +-- destination-address
|       +-- fragment
|       +-- first-fragment
|       +-- protocol
|       +-- source-address
.
.
.
.
```

## 2.3.11 Displaying the state of the configuration

To display the state of the configuration, enter the **info from state** command in candidate or running mode, or the **info** command in state mode.

### Example 1:

To display state information for a specified context from modes that are not state mode:

```
--{ candidate shared default}--[ ]--
# info from state routing-policy policy bgp-export-policy
  routing-policy {
    policy bgp-export-policy {
      statement 999 {
        action {
          accept {
          }
        }
      }
    }
  }
--{ candidate }--[ ]--
```

### Example 2:

To display state information for a specified context from state mode:

```
--{ candidate shared shared default}--[ ]--
# enter state
--{ state }--[ ]--
# info routing-policy policy bgp-export-policy
  routing-policy {
    policy bgp-export-policy {
      statement 999 {
        action {
          accept {
          }
        }
      }
    }
  }
--{ state }--[ ]--
```

## 2.3.12 Executing configuration statements from a file

You can execute configuration statements from a source file consisting of **set** statements such as those generated by the **info flat** command (see [Displaying configuration details](#)). The SR Linux reads the file and executes each configuration statement line-by-line. You can optionally commit the configuration automatically after the file is read.

### Example:

The following example executes a configuration from a specified file:

```
--{ running }--[ ]--
# source config.cfg
Sourcing commands from 'config.cfg'
Executed 20 lines in 1.6541 seconds from file config.cfg
```

Use the **auto-commit** option to commit the configuration after the commands in the source file are executed.

## 2.3.13 Collecting technical support data

Collect technical support data using the **tech-support** command. Use this command in any configuration mode.

### Example:

```
--{ candidate shared default }--[ ]--
# tech-support
Waiting 5 seconds for apps to dump the reports in /tmp/admintech-report-
2019_06_19_20_26_05.zip
Finished collecting in 5s
Admin tech report has been generated at /tmp/admintech-report-
2019_06_19_20_26_05.zip
--{ candidate }--[ ]--
```

You can access the saved file from the bash shell. For example:

```
--{ running }--[ ]--
# bash
[root@3-node_srlinux-A /]# cd /tmp
[root@3-node_srlinux-A tmp]# ls -l
total 6012
-rw-r--r-- 1 root root 6110160 Jun 19 20:26 admintech-report-2019_06_19_20_26_05.zip
[root@3-node_srlinux-A tmp]# exit
logout
--{ running }--[ ]--
```

## 2.4 Formatting and filtering CLI output

Several ways exist to format and filter CLI output. These include:

You can .

- Specifying the display output (text, JSON, or table format)
- Filtering the output using Linux tools such as **grep**
- Using an output filter
- Directing filtered output to a specified file in a specified format

### 2.4.1 Specifying output format

You can display output from SR Linux CLI commands as lines of text, in JSON format, or in a table. By default, output is displayed as lines of text, but you can configure output to be displayed in JSON format by default. See [Configuring the CLI output format](#).

#### Example 1:

The following example displays the output of the **show version** command as lines of text:

```
--{ running }--[ ]--
# show version | as text
-----
Hostname           : 3-node-srlinux-A
Chassis Type       : 7250 IXR-10
Part Number        : Sim Part No.
Serial Number      : Sim Serial No.
System MAC Address : 12:12:02:FF:00:00
Software Version   : v19.11.1
Build Number       : 291-g4664705
Architecture       : x86_64
Last Booted        : 2019-12-07T00:34:48.942Z
Total Memory       : 16396536 kB
Free Memory        : 5321932 kB
-----
```

#### Example 2:

The following example displays the output of the **show version** command in JSON format:

```
--{ running }--[ ]--
# show version | as json
```

```
{
  "basic system info": {
    "Hostname": "3-node-srlinux-A",
    "Chassis Type": "7250 IXR-10",
    "Part Number": "Sim Part No.",
    "Serial Number": "Sim Serial No.",
    "System MAC Address": "12:12:02:FF:00:00",
    "Software Version": "v19.11.1",
    "Build Number": "291-g4664705",
    "Architecture": "x86_64",
    "Last Booted": "2019-12-07T00:34:48.942Z",
    "Total Memory": "16396536 kB",
    "Free Memory": "5319448 kB"
  }
}
```

### Example 3:

The following example displays the output of the **show version** command as a table:

```
--{ running }--[ ]--
# show version | as table
+-----+-----+-----+-----+-----+-----+-----+-----+
| Hostname | Chassis | Part | Serial | System | Software | Architec | Last |
|          | Type   | Number | Number | MAC   | Version  | ture     | Booted |
|          |        |        |        | Address |         |          |        |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 3-node-s | 7250    | Sim Pa | Sim    | 12:12:05 | v19.11.1 | x86_64   | 2019-12- |
| rlinux-A | IXR-10  | rt No. | Serial | :FF:00:0 |          |          | 07T00:34 |
|          |        |        | No.    | 0        |          |          | :48.942Z |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

## 2.4.2 Using Linux output modifiers

Output from SR Linux CLI commands can be piped to standard Linux tools, including **grep**, **more**, **head**, and **tail**.

### Example 1:

The following example pipes the output of the **show interface** command to **grep** so that only lines with **mgmt0** appear in the output:

```
--{ running }--[ ]--
# show interface | grep mgmt0
mgmt0 is up, speed None, type None
  mgmt0.0 is up
```

## Example 2:

The following example pipes the output of the **show interface** command to **more** to display one page of output at a time:

```
--{ running }--[ ]--
# show interface | more
=====
ethernet-1/10 is up, speed 100G, type 100GBASE-CR4 CA-L
  ethernet-1/10.1 is up
    Encapsulation: null
    IPv4 addr    : 192.35.1.0/31 (static)
    IPv6 addr    : 2001:192:35:1::/127 (static, preferred)
    IPv6 addr    : fe80::22e0:9cff:fe78:e2ea/64 (link-layer, preferred)
-----
ethernet-1/21 is up, speed 100G, type 100GBASE-CR4 CA-L
  ethernet-1/21.1 is up
    Encapsulation: null
    IPv4 addr    : 192.45.1.254/31 (static)
    IPv6 addr    : 2001:192:45:1::fe/127 (static, preferred)
    IPv6 addr    : fe80::22e0:9cff:fe78:e2f5/64 (link-layer, preferred)
-----
ethernet-1/22 is up, speed 100G, type 100GBASE-CR4 CA-L
  ethernet-1/22.1 is up
    Encapsulation: null
    IPv4 addr    : 192.45.3.254/31 (static)
    IPv6 addr    : 2001:192:45:3::fe/127 (static, preferred)
    IPv6 addr    : fe80::22e0:9cff:fe78:e2f6/64 (link-layer, preferred)
-----
ethernet-1/3 is up, speed 100G, type 100GBASE-CR4 CA-L
  ethernet-1/3.1 is up
    Encapsulation: null
    IPv4 addr    : 192.57.1.1/31 (static)
    IPv6 addr    : 2001:192:57:1::1/127 (static, preferred)
    IPv6 addr    : fe80::22e0:9cff:fe78:e2e3/64 (link-layer, preferred)
-----
--More--
```

## Example 3:

The following example pipes the output of the **show interface** command to **head** to display only the first 8 lines:

```
--{ running }--[ ]--
# show interface | head --lines 8
=====
ethernet-1/10 is up, speed 100G, type 100GBASE-CR4 CA-L
  ethernet-1/10.1 is up
    Encapsulation: null
    IPv4 addr    : 192.35.1.0/31 (static)
    IPv6 addr    : 2001:192:35:1::/127 (static, preferred)
    IPv6 addr    : fe80::22e0:9cff:fe78:e2ea/64 (link-layer, preferred)
-----
```

## 2.4.3 Using output filters

You can use the **filter** option to limit the output of **show** and **info from state** commands. Filter options include:

- **node** - Defines a specific node to filter on
- **depth** - Filters out sub-nodes that are deeper than the specified depth
- **fields** - Specifies a specific field or fields to filter on
- **keys-only** - Hides all fields
- **non-zero** - Filters integer fields set to 0 and empty strings

The **show** or **info from state** command is piped to the filter command with the following usage: **filter** [<node>] [depth <value>] [fields <value>] [keys-only] [non-zero]

### Example 1:

The following example directs the output of the **info from state** command to filter on the Ethernet node.

```
--{ running }--[ ]--
# info from state interface ethernet-1/1 | filter ethernet
  interface ethernet-1/1 {
    description dut1-dut2-1
    admin-state enable
    mtu 9232
    loopback-mode false
    ifindex 54
    oper-state up
    last-change "an hour ago"
    vlan-tagging false
    ethernet {
      port-speed 10G
      hw-mac-address 00:01:02:FF:00:00
      flow-control {
        receive false
      }
      statistics {
        in-mac-pause-frames 0
        in-oversize-frames 0
        in-jabber-frames 0
        in-fragment-frames 0
        in-crc-error-frames 0
        out-mac-pause-frames 0
        in-64b-frames 0
        in-65b-to-127b-frames 0
        in-128b-to-255b-frames 0
        in-256b-to-511b-frames 0
        in-512b-to-1023b-frames 0
        in-1024b-to-1518b-frames 0
        in-1519b-or-longer-frames 0
        out-64b-frames 0
      }
    }
  }
```

```

        out-65b-to-127b-frames 0
        out-128b-to-255b-frames 0
        out-256b-to-511b-frames 0
        out-512b-to-1023b-frames 0
        out-1024b-to-1518b-frames 0
        out-1519b-or-longer-frames 0
    }
}
}

```

Specify the field option to further filter the output to a specific field. For example:

```

--{ running }--[ ]--
# info from state interface ethernet-1/1 | filter fields ethernet/port-speed
    interface ethernet-1/1 {
        ethernet {
            port-speed 10G
        }
    }
}

```

### Example 2:

The following example shows how you can use the **non-zero** option to display only non-zero values in order to eliminate non-useful information. The use of the non-zero option only filters integer fields that are currently set to 0 and empty strings.

```

--{ running }--[ ]--
# info from state interface ethernet-1/* statistics | filter non-zero
    interface ethernet-1/1 {
        statistics {
            in-octets 1106761
            in-unicast-packets 1592
            in-broadcast-packets 32
            in-multicast-packets 10736
            in-error-packets 5
            out-octets 2859625
            out-unicast-packets 31576
            out-broadcast-packets 18
            out-multicast-packets 243
            carrier-transitions 3
        }
    }
    interface ethernet-1/2 {
        statistics {
            in-octets 1399961
            in-unicast-packets 129
            in-broadcast-packets 11
            in-multicast-packets 14772
            in-error-packets 1
            out-octets 2474023
            out-unicast-packets 26126
            out-broadcast-packets 19
            out-multicast-packets 173
            carrier-transitions 3
        }
    }
}

```

```
interface ethernet-1/20 {
    statistics {
        in-octets 1443012
        in-unicast-packets 350
        in-broadcast-packets 12
        in-multicast-packets 15639
        in-error-packets 3
        out-octets 2319629
        out-unicast-packets 25558
        out-broadcast-packets 17
        out-multicast-packets 199
        carrier-transitions 3
    }
}
```

#### 2.4.4 Directing output to a file

You can direct the output of SR Linux CLI commands to a specified file. The output can be saved as text, in table format, or in JSON format.

**Example:**

The following example directs the output of the **show interface** command to a file. The output is saved in JSON format.

```
--{ running }--[ ]--
# show interface | as json > show interface.json
```

Use `>` to create a new file with the specified filename; if the file already exists, it is replaced. Use `>>` to append the output to the specified file if it exists.

You can access the file using bash mode. For example:

```
--{ * running }--[ ]--
# bash
[bob@3-node-srlinux-A /]# more show_interface.json
{
  "interfaces": [
    {
      "Interface": "ethernet-1/1",
      "subinterfaces": [
        {
          "Subinterface": "ethernet-1/1.1",
          "Oper": "up",
          "IPv4 Addresses": "192.168.11.1/30",
          "IPv6 Addresses": "2001:1::192:168:11:1/126, fe80::1012:5ff:feff:0/64"
        }
      ]
    },
    {

```

## 2.5 Customizing the CLI environment

You can optionally configure the SR Linux CLI environment to change settings such as the command prompt, contents of the bottom toolbar, and the default format for displayed output. You can create aliases for CLI commands. The CLI environment settings can be saved to the default SR Linux configuration or to a specified file and subsequently loaded and applied to the current CLI session.

### 2.5.1 Configuring the CLI prompt

By default, the SR Linux CLI prompt consists of two lines of text, indicating with an asterisk whether the configuration has been modified, the current mode and session type, the current CLI context, and the host name of the SR Linux device, in the following format:

```
--{ modified? mode_and_session_type }--[ context ]--
hostname#
```

#### Example 1:

```
--{ * candidate shared }--[ acl ]--
3-node-srlinux-A#
```

You can configure the SR Linux prompt to include information such as the user name or session ID of the CLI session, the number of changes made to the configuration, and the current local time.

#### Example 2:

The following example adds the local time and session user name to the SR Linux CLI prompt.

```
--{ candidate shared default }--[ ]--
# environment prompt "--{{ {modified}{mode_and_session_type} }}--[ {pwc} ]--{time}--
\n{user}@{host}# "
```

In the example, the local time is configured with the **{time}** keyword, and the session user name is configured with the **{user}** keyword. The line break is configured with **\n**. Use the **environment prompt ?** command to display the keywords that you can configure in the SR Linux CLI prompt.

After you enter this command, the CLI prompt looks like the following:

```
--{ * candidate shared default }--[ acl ]--Wed 03:07PM--
bob@3-node-srlinux-A#
```

## 2.5.2 Configuring the bottom toolbar text

By default, the text that appears at the bottom of the terminal window in SR Linux CLI sessions displays the current mode and session type, whether the configuration has been modified, the user name and session ID of the current AAA session, and the local time, in the following format:

```
Current mode: modified? mode_and_session_type aaa_user (aaa_session_id) time
```

For example:

```
Current mode: * candidate shared                               root (36)  Wed 09:52PM
```

You can configure the bottom toolbar to include information such as the number of changes made to the configuration and the host name.

### Example 1:

The following example adds the number of changes made to the configuration and the host name to the bottom toolbar.

```
--{ candidate shared default }--[ ]--
# environment bottom-
toolbar "Current mode: {modified_with_change_count}{mode_and_session_type} | {aaa_user}@{host} ({aaa_session_id}) {time}"
```

In the example, the number of configuration changes is configured with the **{modified\_with\_change\_count}** keyword, and the host name is configured with the **{host}** keyword. Use the **environment bottom-toolbar ?** command to display the keywords that you can configure in the bottom toolbar.

After you enter this command, the bottom toolbar looks like the following:

```
Current mode: *10 candidate shared                               root@3-node-srlinux-A (36)  Wed 10:18PM
```

## 2.5.3 Configuring the SR Linux CLI engine type

The SR Linux features two versions of the CLI engine: advanced and basic. The advanced CLI engine is enabled by default; it includes the following features:

- Displays a toolbar at the bottom of the terminal window. See [Configuring the bottom toolbar text](#).
- Includes the command auto-complete feature, as described in [Using the CLI auto-complete function](#).

- Allows you to select command options by pressing the Tab key to display the options in a popup, then using the arrow keys to select an option.
- Displays descriptions of command options when you press the ? key.

The basic CLI engine includes a limited set of features compared to the advanced version:

- Omits the bottom toolbar.
- Does not present a selectable list of options when you press the Tab key.
- Displays descriptions of command options when you press the ? key (and press Enter), but only at the # prompt for the current context.

If necessary, you can configure the SR Linux to use the basic CLI engine instead of the advanced CLI engine for CLI sessions.

#### Example:

The following example configures the SR Linux to use the basic CLI engine for CLI sessions:

```
--{ candidate shared default }--[ ]--  
# environment cli-engine type basic
```

## 2.5.4 Configuring the CLI output format

You can configure the output of CLI commands to be displayed as either text or in JSON format.

#### Example:

The following example configures CLI command output to be displayed in JSON format.

```
--{ candidate shared default }--[ ]--  
# environment output-format json
```

Subsequent command output is displayed in JSON format by default. For example:

```
--{ running }--[ ]--  
# show version  
{  
  "basic system info": {  
    "Hostname": "3-node-srlinux-A",  
    "Chassis Type": "7250 IXR-10",  
    "Part Number": "Sim Part No.",  
    "Serial Number": "Sim Serial No.",  
    "System MAC Address": "12:12:02:FF:00:00",  
  }  
}
```

```

    "Software Version": "v19.11.1",
    "Build Number": "291-g4664705",
    "Architecture": "x86_64",
    "Last Booted": "2019-12-07T00:34:48.942Z",
    "Total Memory": "16396536 kB",
    "Free Memory": "5319448 kB"
  }
}

```

## 2.5.5 Configuring CLI command aliases

As a shortcut for entering commands in the CLI, you can configure CLI command aliases. The alias can include one or more CLI command keywords and arguments.

### Example:

The following example configures the alias **display interface** for the SR Linux command **info interface <name> subinterface <index> | as table**:

```

--{ candidate shared default }--[ ]--
# environment alias "display interface" "info /
  interface {} subinterface {subinterface} | as table"

```

In the example, the **display interface** alias consists of the keywords **info interface**, arguments to specify an interface name and subinterface number, and keywords to display the output as a table. The alias name and aliased command are each enclosed in quotes. The arguments are enclosed in braces (**{ }**). The argument **{ }** creates an optional unnamed variable for the interface name, and the argument **{subinterface}** creates an optional parameter named *subinterface*.

When you enter the alias at the CLI prompt, the output of the aliased command is displayed. For example:

```

# display interface ethernet-1/1 subinterface 1
+-----+-----+-----+
|      Interface      | Index | Admin-state |
+=====+=====+=====+
| ethernet-1/1        |      1 | enable      |
+-----+-----+-----+

```

If you omit the optional parameters for the interface and subinterface names, they are treated as wildcards. For example:

```

# display interface
+-----+-----+-----+
|      Interface      | Index | Admin-state |
+=====+=====+=====+
| ethernet-1/1        |      1 | enable      |
| ethernet-1/2        |      1 | enable      |
+-----+-----+-----+

```

|                     |   |        |  |
|---------------------|---|--------|--|
| lo0                 | 1 | enable |  |
| mgmt0               | 0 | enable |  |
| +-----+-----+-----+ |   |        |  |

## 2.5.6 Configuring command auto-completion

By default, if you enter a tab at any mode or level to auto-complete the next command level, a popup appears that displays the available options for that command. See [Using the CLI auto-complete function](#).

You can optionally configure the space bar to provide the same function as the Tab key, so that pressing the space bar auto-completes the command.

### Example:

The following example configures the space bar to auto-complete commands in SR Linux CLI sessions:

```
--{ candidate shared default }--[ ]--
# environment complete-on-space true
```

## 2.5.7 Displaying the CLI environment configuration

To display the CLI environment configuration, including any CLI command aliases, use the **environment show** command.

### Example:

```
--{ candidate shared default }--[ ]--
# environment show
[alias]
"show system configuration session" = "info from state /
system configuration session {} | as table | filter fields username type started"
"show system configuration checkpoint" = "info from state /
system configuration checkpoint {} | as table | filter fields name comment username
created"
"display interface" = "info / interface {} subinterface {subinterface} | as table

[bottom-toolbar]
value = "Current mode: {modified}{mode_and_session_type} | {aaa_user} ({aaa_session_
id}) {time}"

[cli-engine]
type = "advanced"

[output-display-format]
value = "text"
```

```
[prompt]
value = "--{{ {modified}{mode_and_session_type} }}--[ {pwc} ]--\n{host}# "

[space-completion]
enabled = false
```

## 2.5.8 Managing CLI environment settings

You can save the current CLI environment settings to the default SR Linux configuration or to a file, and you can load CLI environment settings from the default SR Linux configuration or from a file.

### Example 1:

The following example saves the current CLI environment settings to the default SR Linux configuration:

```
--{ candidate shared default}--[ ]--
# environment save home
Saved configuration to /root/.srlinuxrc
```

### Example 2:

The following example saves the current CLI environment settings to a file:

```
--{ candidate shared default}--[ ]--
# environment save file env.cfg
Saved configuration to env.cfg
```

### Example 3:

The following example loads CLI environment settings from the default SR Linux configuration:

```
--{ candidate shared default}--[ ]--
# environment load home
Loaded configuration from /root/.srlinuxrc
```

### Example 4:

The following example loads CLI environment settings from a file:

```
--{ candidate shared default}--[ ]--
# environment load file env.cfg
Loaded configuration from env.cfg
```

## 3 RPC overview and supporting interfaces

SR Linux supports the gNMI and JSON interfaces that use the Remote Procedure Call (RPC) protocol for the modification and retrieval of a configuration from a target device as if it were a local object. This chapter provides a general RPC overview and defines how SR Linux implements the gNMI and JSON serving RPCs.

### 3.1 RPC overview

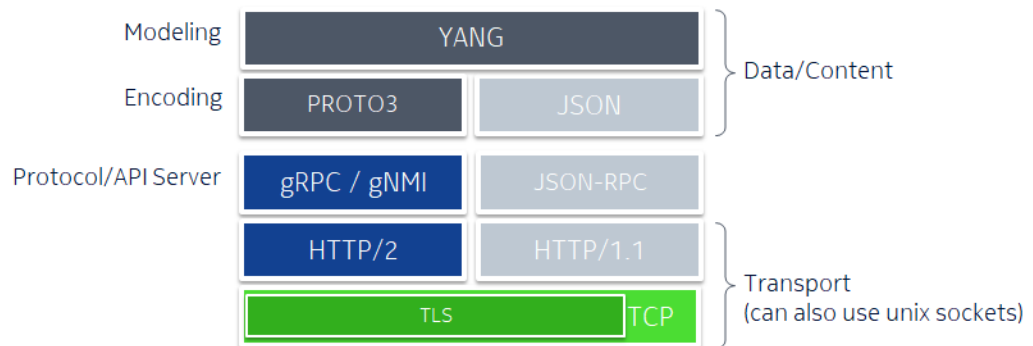
An RPC executes a procedure or method on a remote device in a way similar to one executed locally. Using an RPC should look and feel like a local procedure call and achieve similar results.

When RPCs operate in a server/client model, the client executes an application and requests some method be executed on a server. The server receives the RPC, executes the method, and sends the outcome back to the server. For this to occur, the following are needed:

- A server application that contains a set of methods (or service) which a client can call. For example, a gNMI server or JSON-RPC server. Each needs to receive an RPC from a client for a specific method or service, execute the requested methods/service, and return data.
- A common way of describing which method to execute and associated data. This is referred to as encoding/decoding or serialization/deserialization. For example, proto buffers or JSON.
- A transport mechanism between the two devices. In the case of gNMI and JSON-RPC this is done using HTTP/2 and HTTP1.1 over TCP secured by TLS.

For RPCs to relate to network applications and a network operating system, there needs to be a way to define the data model of the network constructs which an RPC must perform. In the SR Linux, all applications are modeled in YANG. See [Figure 1](#).

**Figure 1 RPC-based interfaces in SR Linux**



### 3.1.1 gNMI path convention

Because SR Linux is modeled in YANG, for RPCs to retrieve or configure an SR Linux device (set) the data location or path to the data within the YANG model must be specified to the RPC. For example, to configure a description under a BGP peer, the RPC server must be able to reference this specific data and its location in the SR Linux data model.

Both the JSON-RPC and gNMI use the gNMI path convention to describe the location or path in SR Linux. The gNMI path convention is commonly referred to as the “path” and defines the data structure of a path to reference a config or state leaf within SR Linux.

A path is an ordered list of path elements with each element containing an element name and one or more key value pairs associated with the element. For example:

```
path: <
  elem: <
    name: "interface"
    key: <
      key: "name"
      value: "ethernet-1/20"
    >
  >
  elem: <
    name: "subinterface"
    key: <
      key: "index"
      value: "1"
    >
  >
```

A path can also be displayed as a string which is more human readable. The rules for building this string type are:

- Each path element is separated by a “/” character and listed in sequence.
- A path element which contains one or more key/value pairs is represented by the path element name followed the key/value wrapped in brackets ( [ ] ).
- The root of the data tree is represented by a single “/”.

The following is a human readable path for the previous example:

```
/interface=ethernet-1/20/subinterface[index=1]
```

## 3.2 Configuring a gNMI or JSON server

The SR Linux can enable a gNMI server that allows external gNMI clients to connect to the device and modify the configuration and collect state information. You can also enable a JSON-RPC server on the SR Linux device, which lets you issue JSON-formatted requests to the device to retrieve and set configuration and state.

Refer to the Management Servers chapter in the *SR Linux Configuration Basics Guide* for details on how to configure these servers.

## 4 gNMI interface

A gRPC Network Management Interface (gNMI) is a gRPC based protocol that defines a service or set of services or RPC methods used to configure and retrieve data from network devices.

The SR Linux provides a gNMI-based RPC for the modification and retrieval of a configuration. Supported RPCs are:

- get
- set
- subscribe
- capabilities

### 4.1 Common notification messages

When the SR Linux gNMI process communicates data to a client, it uses common notification messages. Notification messages use the fields shown in [Table 6](#).

**Table 6** Common notification fields

| Field     | Definition                                                                                                                                                        |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| timestamp | Time data was collected                                                                                                                                           |
| prefix    | Prefix applied to all path fields included in the notification message. The paths expressed within the message are formed by the concatenation of prefix + path.  |
| update    | List of update messages that indicate changes in the underlying data. Subfields are: <ul style="list-style-type: none"><li>• path</li><li>• val (value)</li></ul> |
| delete    | List of paths indicating the deletion of data nodes                                                                                                               |

#### 4.1.1 Timestamps

Timestamp values are represented in nanoseconds. The value is encoded as a signed 64-bit integer (int64).

## 4.1.2 Path prefix

A prefix can be specified to reduce the lengths of path fields within a message. The absolute path is a concatenation of the path elements representing the prefix and the list of path elements in the path field. For example:

```
notification: <
timestamp: (timestamp) // timestamp as int64
prefix: <
  elem: <
    name: "a"
  >
  elem: <
    name: "b"
    key: <
      key: "name"
      value: "b1"
    >
  >
  elem: <
    name: "c"
  >
>
update: <
  path: <
    elem: <
      name: "d"
    >
  >
  value: <
    val: <
      json_val: "AStringValue"
    >
  >
>
update: <
  path: <
    elem: <
      name: "e"
    >
  >
  val: <
    json_val: 10042 // converted to int representation
  >
>
>
```

## 4.1.3 Paths

Paths are represented according to gNMI path conventions. Each path is represented by an ordered list of PathElem messages, starting at the root node, and ending at the most specific path element (versus a single string with a "/" character separating each element). Each PathElem message contains the name of the node within the data tree, along with any associated keys and attributes that may be required. For example:

```
path: <
  elem: <
    name: "a"
  >
  elem: <
    name: "e"
    key: <
      key: "key"
      value: "k1"
    >
  >
  elem: <
    name: "f"
  >
  elem: <
    name: "g"
  >
>
```

Multiple paths are supported. Multiple notification messages are triggered in response to each path. For example:

```
path <
  elem <
    name: "interface"
    key <
      key: "name"
      value: "mgmt0"
    >
  >
>
path <
  elem <
    name: "system"
  >
>
type: 1
encoding: JSON_IETF
```

## 4.1.4 Data node values

The value of a data node can be the following:

- scalar types, such as a string in the `string_val` field, `int64` in the `int_val` field, `uint64` in the `uint_val` field, `bool` in the `bool_val` field, `bytes`, and `float` in the `float_val` field
- additional types used in some schema languages, such as `decimal64` in the `decimal_val` field and `ScalarArray` in the `leaflist_val` field
- structured data types

### 4.1.4.1 Structured data types

When structured data is sent in an update message, it is serialized according to supported encoding, as shown in [Table 7](#).

**Table 7** Encoding for structured data

| Data type | Description                                                        | Field                      |
|-----------|--------------------------------------------------------------------|----------------------------|
| ASCII     | An ASCII encoded string                                            | <code>ascii_val</code>     |
| JSON_IETF | A JSON encoded string using JSON encoding compatible with RFC 7951 | <code>json_ietf_val</code> |

## 4.2 gNMI get RPC

The get RPC lets you obtain a view of the existing state. A `GetRequest` message is sent to the target (SR Linux gNMI process `gnmi_mgr`) that specifies the data to retrieve. A `GetResponse` message is returned that reflects the values of specified leafs at the collection time.

The get RPC is recommended for retrieving small data sets. For larger data sets, the [gNMI subscribe RPC](#) is recommended, using the ONCE mode.

## 4.2.1 GetRequest message

A GetRequest message retrieves a view of data from the server. A Get RPC requests the server retrieve a subset of the data tree as specified by the paths included in the message and serializes this using the specified encoding.

The GetRequest message uses the fields shown in [Table 8](#).

**Table 8** GetRequest fields

| Field     | Definition                                                                                                                                                                               |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| path      | Path (or set of paths) for the requested data view. Wildcards are permitted.                                                                                                             |
| type      | Type of data requested. Supported options are: <ul style="list-style-type: none"><li>• CONFIG (configurable read/write data)</li><li>• STATE (non-configurable read-only data)</li></ul> |
| encoding  | Encoding that the target should use (ASCII or JSON_IETF). If not specified, JSON is the default.                                                                                         |
| extension | Repeated field to carry gNMI extensions                                                                                                                                                  |

## 4.2.2 GetResponse message

The GetResponse message uses the fields shown in [Table 9](#).

**Table 9** GetResponse fields

| Field        | Definition                                                                                                                                                   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| notification | Set of notification messages for each path specified in the GetRequest; see <a href="#">Common notification messages</a> for definitions of common messages. |
| extension    | Repeated field to carry gNMI extensions                                                                                                                      |

## 4.3 gNMI set RPC

The set RPC lets you modify an existing state. A SetRequest message is sent to the target (SR Linux gNMI process gnmi\_mgr) that specifies the required modifications. The server deletes, replaces, and updates paths based on the order they are listed. For each operation designated in the SetRequest message, an UpdateResult message is included in the SetResponse message.

### 4.3.1 SetRequest message

The SetRequest message uses the fields shown in [Table 10](#).

**Table 10** SetRequest fields

| Field     | Definition                                                           |
|-----------|----------------------------------------------------------------------|
| prefix    | A specified prefix is applied to all defined paths within each field |
| delete    | A set of paths to be removed from the data tree                      |
| replace   | A set of update messages that defines content to replace             |
| update    | A set of update messages that defines content to update              |
| extension | Repeated field to carry gNMI extensions                              |

An update message indicates changes to paths where a new value is required. Update messages contain the following:

- path - the path of the element to be modified
- value - a value to apply to the specified node

All changes to the state included in a SetRequest message are considered part of a transaction. Either all modifications are applied or changes are rolled back to reflect the original state. For changes to be applied together, they must be in a single SetRequest message.

For replace operations, the behavior of omitted data elements depends on whether they are non-default values (set by a previous SetRequest message) or unmodified defaults. When the replace operation omits values that have been previously set, they are deleted from the data tree. Otherwise, omitted data elements are created with their default values.

For update operations, only the value of the data elements explicitly specified are changed.

## 4.3.2 SetResponse message

The SetResponse message uses the fields shown in [Table 11](#).

**Table 11** SetResponse fields

| Field     | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prefix    | The prefix specified for all paths                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| response  | A list of responses (one per operation). Each response consists of an UpdateResult message with the following: <ul style="list-style-type: none"><li>• timestamp - time when the SetRequest message was accepted</li><li>• path - path defined in SetRequest message. A prefix may be present to reduce repetition of path elements.</li><li>• op - the operation performed on the path (delete, replace, or update)</li><li>• message - a status message</li></ul> |
| extension | Repeated field to carry gNMI extensions                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 4.4 gNMI subscribe RPC

The subscribe RPC lets you receive updates relating to the state of data instances. The user creates a subscription using the subscribe RPC with the desired subscription mode. The defined mode triggers how and when the data is sent to the client.

A SubscribeRequest message is sent to the target (SR Linux gNMI process `gnmi_mgr`) to request updates for one or more paths. A SubscribeResponse message is sent to the client over an established RPC.

## 4.4.1 SubscribeRequest message

The SubscribeRequest message uses the fields shown in [Table 12](#).

**Table 12** SubscribeRequest fields

| Field     | Definition                                                               |
|-----------|--------------------------------------------------------------------------|
| subscribe | A SubscriptionList message specifying a new set of paths to subscribe to |
| extension | Repeated field to carry gNMI extensions                                  |

Subscriptions are set once and cannot be modified. A new subscribe RPC call must be created for new paths. To end an existing subscription, the client must cancel the subscribe RPC that relates to the subscription.

### 4.4.1.1 SubscriptionList message

A SubscriptionList message indicates a set of paths where common subscription behavior is required. The SubscriptionList message uses the fields shown in [Table 13](#).

**Table 13** SubscriptionList fields

| Field        | Definition                                                                                                                                                                                                                                                                                           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| subscription | A set of subscription messages indicating the paths associated with the subscription                                                                                                                                                                                                                 |
| mode         | Type of subscription to create: <ul style="list-style-type: none"> <li>• ONCE</li> <li>• STREAM (default) <ul style="list-style-type: none"> <li>– ON_CHANGE</li> <li>– SAMPLE <ul style="list-style-type: none"> <li>• sample_interval</li> </ul> </li> <li>– TARGET_DEFINED</li> </ul> </li> </ul> |
| extension    | Repeated field to carry gNMI extensions                                                                                                                                                                                                                                                              |

ONCE subscriptions are one-time requests. A ONCE subscription is created by sending a SubscribeRequest message with the subscribe field containing a SubscriptionList, with the mode type set to ONCE. The relevant update messages are sent and the RPC channel is closed.

STREAM subscriptions are long-lived and transmit updates indefinitely. A STREAM subscription is created by sending a SubscribeRequest message with the subscribe field containing a SubscriptionList, with the mode type set to STREAM. The STREAM mode subscription message also specifies a mode.

- **ON\_CHANGE** - Data updates are only sent when the value of the data item changes.
- **SAMPLE** - Data is sent at specified intervals as specified in the sample\_interval field. The maximum sample rate is a 64-bit integer in nanoseconds and minimum is 0.
- **TARGET\_DEFINED** - The target determines the best subscription type to create on a per-leaf basis. For example, if the path specified refers to leaves that are event-driven, then an ON\_CHANGE subscription may be created. If the data represents counters values, a SAMPLE subscription may be created.

## 4.4.2 SubscribeResponse message

The SubscribeResponse message uses the fields shown in [Table 14](#).

**Table 14** SubscribeResponse fields

| Field                                 | Definition                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| update<br><br>OR<br><br>sync_response | A response field. Only one type can be specified per message: <ul style="list-style-type: none"> <li>• update - message providing an update value for a subscribed data entity</li> <li>• sync_response - a Boolean field indicating that all data values corresponding to the paths have been transmitted at least once (not used with ONCE mode subscriptions)</li> </ul> |
| extension                             | Repeated field to carry gNMI extensions                                                                                                                                                                                                                                                                                                                                     |

## 4.5 gNMI capabilities RPC

The capabilities RPC lets you discover the capabilities of a specific gNMI server.

A CapabilityRequest message is sent by the client to request capability information from the target. The target replies with a CapabilityResponse message that includes its gNMI service version, the versioned data models it supports, and the supported data encodings.

This information is used in subsequent RPC messages from the client to indicate the set of models that the client will use, and the encoding used for data.

### 4.5.1 CapabilityRequest message

The CapabilityRequest message is sent by the client to request capability information from the target. The CapabilityRequest message carries a single repeated extension field which can be used to carry gNMI extensions.

```
message CapabilityRequest {  
    repeated gnmi_ext.Extension extension = 1;  
}
```

### 4.5.2 CapabilityResponse message

A CapabilityResponse message is sent from the target and includes the following fields:

- supported\_models - a set of ModelData messages describing each model supported by the target
- supported\_encodings - an enumerated field describing the data encodings supported by the target (ASCII and JSON\_IETF are supported)
- gNMI\_version - the version of the gNMI service supported by the target
- encoding - a repeated field for gNMI extensions

```
message CapabilityResponse {  
    repeated ModelData supported_models = 1;  
    repeated Encoding supported_encodings = 2;  
    string gNMIversion = 3;  
    repeated gnmi_ext.Extension extension = 4;  
}
```

## 4.6 Candidate mode

The gNMI uses its own private candidate that allows multiple users or services to make simultaneous changes to a configuration.

## 4.7 gNMI examples

Open source clients can be used to run GetRequests, SetRequests, subscriptions, and capabilities. The examples that follow show requests and responses using the following clients although any client that conforms to gNMI specifications can be used:

- `gnmi_get` — used for simple GetRequests
- `gnmi_set` — used for simple SetRequests
- `gnmi_cli` — used for SubscribeRequests, and advanced GetRequests and SetRequests
- `gnmi_capabilities` — used for CapabilityRequests

### 4.7.1 `gnmi_get` examples

The `get` gNMI-RPC lets you retrieve state and configuration from a datastore. The following examples are shown:

- Example 1: shows a get all request
- Example 2: shows a get all response
- Example 3: shows a get interface with wildcard key request
- Example 4: shows a get interface with wildcard key response

#### Example 1: get all request

```
# gnmi_get -target_addr 172.18.0.6:50052 -insecure -xpath '/'  
== getRequest:  
path: <  
>  
encoding: JSON_IETF
```

#### Example 2: get all response

```
notification: <  
  timestamp: 1565672122888042050
```

```
update: <
  path: <
  >
  val: <
    json_ietf_val: "{\n \"srl_nokia-acl:acl\": {\n \"ipv4-
    filter\" ---- snip ---- }\n }\n\n"
  >
  >
  >
```

```
# gnmi_get -target_addr 172.18.0.6:50052 -insecure -xpath
'/interface[name=mgmt0]/subinterface[index=*]'
== getRequest:
```

```
path: <
  elem: <
    name: "interface"
    key: <
      key: "name"
      value: "mgmt0"
    >
  >
  elem: <
    name: "subinterface"
    key: <
      key: "index"
      value: "*"
    >
  >
  >
encoding: JSON IETF
```

```
notification: <
  timestamp: 1565671919030747121
  update: <
    path: <
      elem: <
        name: "srl_nokia-interfaces:interface"
        key: <
          key: "name"
          value: "mgmt0"
        >
      >
    >
  val: <
    json_ietf_val: "{\n  \"name\": \"mgmt0\", \n  \"subinterface\":\n  [\n    {\n      \"index\": 0, \n      \"admin-state\": \"enable\", \n\n\n      \"ip-mtu\": 1500, \n      \"ifindex\": 524288000, \n      \"operstate\":\n      \"up\", \n      \"last-change\": \"2019-08-11T17:21:48.366Z\", \n      \"ipv4\": {\n        \"allow-directedbroadcast\":\n        false, \n        \"dhcp-client\": true, \n\n        \"address\": [\n          {\n            \"ip-prefix\": in\n            \"172.18.0.6/24\", \n            \"origin\": \"dhcp\" \n          }\n        ], \n        \"srl nokia-interfaces-nbr:arp\": {\n
```

## 2 gnmi\_set examples

- Example 1: shows a set delete request
- Example 2: shows a set delete response
- Example 3: shows a set update all request
- Example 4: shows a set update all response

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```
    elem: <
      name: "name"
    >
    elem: <
      name: "host-name"
    >
  >
```

### Example 2: set a delete response

```
response: <
  path: <
    elem: <
      name: "system"
    >
    elem: <
      name: "name"
    >
    elem: <
      name: "host-name"
    >
  >
  op: DELETE
>
timestamp: 1567203341816078044
```

### Example 3: set a update all request

```
# gnmi_set -target_addr 172.18.0.3:50052 -username admin -password
admin -insecure -update /system/name/host-name:replaced-host -replace
/system/name/domain-name:replaced-domain
== setRequest:
replace: <
  path: <
    elem: <
      name: "system"
    >
    elem: <
      name: "name"
    >
    elem: <
      name: "domain-name"
    >
  >
  val: <
    string_val: "replaced-domain"
  >
>
update: <
  path: <
    elem: <
      name: "system"
    >
    elem: <
      name: "name"
    >
    elem: <
```

```
        name: "host-name"
      >
    >
    val: <
      string_val: "replaced-host"
    >
  >
```

#### Example 4: set a update all response

```
response: <
  path: <
    elem: <
      name: "system"
    >
    elem: <
      name: "name"
    >
    elem: <
      name: "domain-name"
    >
  >
  op: REPLACE
response: <
  path: <
    elem: <
      name: "system"
    >
    elem: <
      name: "name"
    >
    elem: <
      name: "host-name"
    >
  >
  op: UPDATE
>
timestamp: 1567204165851469784
```

## 4.7.3 gnmi\_cli examples

The **cli** gNMI-RPC lets you subscribe and receive updates on the state of a data instance. The following examples are shown:

- Example 1: Subscribe - ONCE for all (one-time subscription) request
- Example 2: Subscribe - ONCE for all (one-time subscription) response
- Example 3: Subscribe - STREAM ON\_CHANGE interface (long term subscription) request
- Example 4: Subscribe - STREAM ON\_CHANGE interface (long term subscription) response

In these examples, **-qt** specifies the subscription type. ONCE mode is the default and therefore is not shown in the first example.

### Example 1: Subscribe ONCE for all request

```
# gnmi_cli -a 172.18.0.6:50052 -insecure -q '/'
```

### Example 2: Subscribe ONCE for all response

```
{
  "acl": {
    "ipv4-filter": {
      "allow_sip_dip": {
        "entry": {
          "10": {
            "action": {
              "accept": {
                "log": "false"
              }
            }
          }
        }
      }
    }
  }
}
-- Snip -
```

### Example 3: Subscribe STREAM ON\_CHANGE interface request

```
# gnmi_cli -a 172.18.0.6:50052 -insecure --qt streaming -q
'/interface[name=mgmt0]'
```

### Example 4: Subscribe STREAM ON\_CHANGE interface response

```
{
  "interface": {
    "mgmt0": {
      "admin-state": "enable",
      "ethernet": {
        "flow-control": {
          "receive": "false"
        }
      },
      "hw-mac-address": "02:42:AC:12:00:06",
    }
  }
}
```

```

    "statistics": {
      "in-crc-errors": "0",
      "in-fragment-frames": "0",
      "in-jabber-frames": "0",
      "in-mac-pause-frames": "0",
      "in-oversize-frames": "0",
      "out-mac-pause-frames": "0"
    }
  },
  "ifindex": "524304383",
  "last-change": "2019-08-30T18:44:45.490Z",
  "mtu": "1514",
  "oper-state": "up",
  "statistics": {
    "carrier-transitions": "1",
    "in-broadcast-pkts": "5",
    "in-errors": "0",
    "in-fcs-errors": "0",
    "in-multicast-pkts": "1356",
    "in-octets": "612022",
    "in-unicast-pkts": "4662",
    "out-broadcast-pkts": "1",
    "out-errors": "0",
    "out-multicast-pkts": "456",
    "out-octets": "2724476",
    "out-unicast-pkts": "5505"
  },
  "subinterface": {
    "0": {
      "admin-state": "enable",
      "ifindex": "524288000",
      "ip-mtu": "1500",
      "ipv4": {
        "address": {
          "172.18.0.6/24": {
            "origin": "dhcp"
          }
        },
        "allow-directed-broadcast": "false",
        "arp": {
          "neighbor": {
            "172.18.0.1": {
              "expiration-time": "2019-08-31T01:13:22.987Z",
              "link-layer-address": "02:42:45:9D:DB:FC",
              "origin": "dynamic"
            },
            "172.18.0.2": {
              "expiration-time": "2019-08-30T22:44:54.422Z",
              "link-layer-address": "02:42:AC:12:00:02",
              "origin": "dynamic"
            }
          }
        },
        "timeout": "14400"
      },
      "dhcp-client": "true"
    },
    "ipv6": {
      "address": {
        "2001:172:18::6/80": {

```

```

        "origin": "dhcp",
        "status": "preferred"
    },
    "fe80::42:acff:fe12:6/64": {
        "origin": "link-layer",
        "status": "preferred"
    }
},
"dhcp-client": "true",
"neighbor-discovery": {
    "dup-addr-detect": "true",
    "reachable-time": "30",
    "stale-time": "14400"
}
},
"last-change": "2019-08-30T18:44:45.490Z",
"oper-state": "up",
"qos": {
    "input": {
        "classifiers": {
            "ipv4-dscp": "default",
            "ipv6-dscp": "default",
            "mpls-tc": "default"
        }
    }
},
"statistics": {
    "in-discarded-pkts": "0",
    "in-error-pkts": "0",
    "in-forwarded-octets": "0",
    "in-forwarded-pkts": "0",
    "in-octets": "404380",
    "in-pkts": "4679",
    "in-terminated-octets": "404380",
    "in-terminated-pkts": "4679",
    "out-discarded-pkts": "0",
    "out-error-pkts": "0",
    "out-forwarded-octets": "2409995",
    "out-forwarded-pkts": "5511",
    "out-octets": "2409995",
    "out-pkts": "5511"
}
}
},
"vlan-tagging": "false"
}
}
{
    "interface": {
        "mgmt0": {
            "statistics": {
                "in-octets": "615366"
            }
        }
    }
}
}
{
    "interface": {
        "mgmt0": {

```

```

        "statistics": {
            "in-unicast-pkts": "4693"
        }
    }
}
{
    "interface": {
        "mgmt0": {
            "statistics": {
                "out-octets": "2736287"
            }
        }
    }
}
.
.
.

```

## 4.7.4 gnmi\_capabilities example

The **capabilities** gNMI-RPC lets you discover the capabilities of a specific gNMI server. The following example is shown:

- Example 1: shows a request to obtain model, data encodings, and version for a specified server.
- Example 2: shows a response for a request (output from Example 1).

### Example 1: Request server capabilities for specified server

```
gnmi_capabilities -username admin -password admin --target_addr [172.18.0.8]:50264
```

### Example 2: Response to a server capabilities request

```

supported_models: <
  name: "urn:srl_nokia/aaa:srl_nokia-aaa"
  organization: "Nokia"
  version: "2020-12-31"
>

supported_models: <
  name: "urn:srl_nokia/aaa-types:srl_nokia-aaa-types"
  organization: "Nokia"
  version: "2019-11-30"
>
.
.
.
supported_encodings: JSON_IETF
supported_encodings: ASCII
supported_encodings: 45
supported_encodings: 44

```

---

```
supported_encodings: 46
supported_encodings: 47
gNMI_version: "0.7.0"
```

## 5 JSON interface

The SR Linux provides a JSON-based Remote Procedure Call (RPC) for both CLI commands and configuration. The JSON API allows the operator to retrieve and set the configuration and state, and provide a response in JSON format. This JSON-RPC API models the CLI implemented on the system.

If output from a command cannot be displayed in JSON, the text output is wrapped in JSON to allow the application calling the API to retrieve the output. During configuration, if a TCP port is in use when the JSON-RPC server attempts to bind to it, the commit fails. The JSON-RPC supports both normal paths, as well as XPATHs.

### 5.1 JSON message structure

The JSON RPC requires a specific message structure. The jsonrpc version, id, method, and params are required in all requests.

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "get",
  "params": {
  }
}
```

Within these required elements can be additional mandatory and conditional elements; see [Table 15](#).

**Table 15** Required JSON request structure

| Required request elements | Description                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| jsonrpc                   | Version, which must be "2.0". No other JSON RPC versions are currently supported.                                                                           |
| id                        | Client-provided integer. The JSON RPC responds with the same ID, which allows the client to match requests to responses when there are concurrent requests. |
| method                    | Defines the method accessed with the JSON RPC. Supported options are get, set, and cli. See <a href="#">method options</a> for more information.            |

**Table 15 Required JSON request structure (Continued)**

| Required request elements | Description                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| params                    | Defines a container for any parameters related to the request. The type of parameter is dependent on the method used. See <a href="#">params options</a> for more information. |

## 5.1.1 method options

[Table 16](#) defines supported JSON RPC method options.

**Table 16 JSON RPC method options**

| Method option | Description                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| get           | Used to retrieve configuration and state details from the system. The get method can be used with candidate, running, and state datastores, but cannot be used with the tools datastore.                          |
| set           | Used to set a configuration or run operational transaction. The set method can be used with the candidate and tools datastores.                                                                                   |
| validate      | Used to verify that the system will accept a configuration transaction before applying it to the system.                                                                                                          |
| cli           | Used to run CLI commands. The get and set methods are restricted to accessing data structures via the YANG models, but the cli method can access any commands added to the system via python plug-ins or aliases. |

## 5.1.2 params options

Table 17 defines valid JSON RPC params options.

**Table 17** JSON RPC params options

| params option                                                                                                                                                                                                                                                                                                                                                                                       | Descriptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>commands</b> - Mandatory. List of commands used to execute against the called method. Multiple commands can be executed with a single request. Supported commands are: <ul style="list-style-type: none"> <li>• <b>action</b></li> <li>• <b>path</b></li> <li>• <b>path-keywords</b></li> <li>• <b>datastore</b></li> <li>• <b>recursive</b></li> <li>• <b>include-field-defaults</b></li> </ul> | <b>action</b> command - Conditional mandatory; used with the set and validate methods. Supported options are: <ul style="list-style-type: none"> <li>• <b>replace</b> — Replaces the entire configuration within a specific context with the supplied configuration; equivalent to a delete/update.</li> <li>• <b>update</b> — Updates a leaf or container with the specified value.</li> <li>• <b>delete</b> — Deletes a leaf or container. All children beneath the parent are removed from the system.</li> </ul> <p>Note: When the action command is used with the tools datastore, update is the only supported option.</p>                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                     | <b>path</b> command - Mandatory with the get, set and validate methods. This value is a string that follows the gNMI path specification <sup>1</sup> in human-readable format: <ul style="list-style-type: none"> <li>• “/” separates nodes</li> <li>• keys are specified using [&lt;key-name&gt;=&lt;key-value&gt;]</li> <li>• if key-value contains “]”, it must be escaped (“\” before the “]”)</li> <li>• fields end with “.” or “,value&gt;”</li> </ul> <p>Example: /interface[name=mgmt0]</p>                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                     | <b>path-keywords</b> command - Optional; used to substitute named parameters with the path field. More than one keyword can be used with each path.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                                     | <b>datastore</b> command - Optional; selects the datastore to perform the method against. Supported options are: <ul style="list-style-type: none"> <li>• <b>candidate</b> — Used to change the configuration of the system with the get, set, and validate methods; default datastore is used if the datastore parameter is not provided.</li> <li>• <b>running</b> — Used to retrieve the active configuration with the get method.</li> <li>• <b>state</b> — Used to retrieve the running (active) configuration along with the operational state.</li> <li>• <b>tools</b> — Used to perform operational tasks on the system; only supported with the update action command and the set method.</li> </ul> |

**Table 17 JSON RPC params options (Continued)**

| params option                                                                                                                                                                                                                                 | Descriptions                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>commands</b><br>(cont.)                                                                                                                                                                                                                    | <b>recursive</b> command - Optional; a Boolean used to retrieve children underneath the specific path. The default = true.                                                                          |
|                                                                                                                                                                                                                                               | <b>include-field-defaults</b> command - Optional; a Boolean used to show all fields, regardless if they have a directory configured or are operating at their default setting. The default = false. |
| <b>output-format</b> - Optional.<br>Defines the output format as: <ul style="list-style-type: none"> <li>• json</li> <li>• text</li> <li>• table</li> </ul>                                                                                   | Output defaults to JSON if not specified.                                                                                                                                                           |
| Table Note 1: gNMI path specification reference:<br><a href="https://github.com/openconfig/reference/blob/master/rpc/gnmi/gnmi-path-conventions.md">https://github.com/openconfig/reference/blob/master/rpc/gnmi/gnmi-path-conventions.md</a> |                                                                                                                                                                                                     |

## 5.2 JSON responses

The JSON RPC returns one entry for each command that was executed. For methods that contain non-response (other than acknowledging the command), a response is returned, but with an empty list.

When the cli method is used, each executed command returns an individual response.

## 5.3 Candidate mode

JSON uses its own private candidate that allows multiple users or services to make simultaneous changes to a configuration.

## 5.4 Logical expressions

For logical expressions, support is provided for the asterisk (“\*”) which can be used to reference all keys within a list.

## 5.5 JSON examples

The following provides JSON examples (both requests and responses) for these method options:

- [JSON get example](#)
- [JSON set example](#)
- [JSON delete example](#)
- [JSON validate example](#)
- [JSON CLI example](#)

### 5.5.1 JSON get example

The **get** method lets you retrieve configuration and state details from the system.

- Example 1: shows a get method using the path, datastore, and recursive commands with the recursive option set to true (to retrieve children underneath the specific path).
- Example 2: shows a multiple get request where multiple commands are executed with a single request.

#### Example 1: Single get with recursive request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "get",
  "params": {
    "commands": [
      {
        "path": "/interface[name=mgmt0]",
        "datastore": "state",
        "recursive": true
      }
    ]
  }
}
```

#### Response (single get with recursive)

```
{
  "result": [
    {
      "name": "mgmt0",
      "admin-state": "enable",
      "mtu": 1514,

```

```
"ifindex": 524304383,
"oper-state": "up",
"last-change": "2019-07-12T16:53:39.291Z",
"statistics": {
  "in-octets": "4545395",
  "in-unicast-pkts": "1178",
  "in-broadcast-pkts": "130",
  "in-multicast-pkts": "27560",
  "in-discards": "0",
  "in-errors": "0",
  "in-unknown-protos": "0",
  "in-fcs-errors": "0",
  "out-octets": "1735990",
  "out-unicast-pkts": "1125",
  "out-broadcast-pkts": "38",
  "out-multicast-pkts": "9187",
  "out-discards": "0",
  "out-errors": "0",
  "carrier-transitions": "1"
},
"ethernet": {
  "hw-mac-address": "02:42:AC:12:00:05",
  "statistics": {
    "in-mac-control-frames": "0",
    "in-mac-pause-frames": "0",
    "in-oversize-frames": "0",
    "in-jabber-frames": "0",
    "in-fragment-frames": "0",
    "in-crc-errors": "0",
    "out-mac-control-frames": "0",
    "out-mac-pause-frames": "0"
  }
},
"subinterface": [
  {
    "index": 0,
    "admin-state": "enable",
    "ip-mtu": 1500,
    "ifindex": 524288000,
    "oper-state": "up",
    "last-change": "2019-07-12T16:53:39.291Z",
    "ipv4": {
      "allow-directed-broadcast": false,
      "dhcp-client": true,
      "address": [
        {
          "ip-prefix": "172.18.0.5/24",
          "origin": "dhcp"
        }
      ]
    },
    "srl_nokia-interfaces-nbr:arp": {
      "timeout": 14400,
      "neighbor": [
        {
          "ipv4-address": "172.18.0.1",
          "link-layer-address": "02:42:90:06:D7:26",
          "origin": "dynamic",
          "expiration-time": "2019-07-15T21:37:28.987Z"
        }
      ]
    }
  }
],
```

```

    {
      "ipv4-address": "172.18.0.2",
      "link-layer-address": "02:42:AC:12:00:02",
      "origin": "dynamic",
      "expiration-time": "2019-07-16T00:44:17.673Z"
    },
    {
      "ipv4-address": "172.18.0.3",
      "link-layer-address": "02:42:AC:12:00:03",
      "origin": "dynamic",
      "expiration-time": "2019-07-16T01:20:48.600Z"
    },
    {
      "ipv4-address": "172.18.0.4",
      "link-layer-address": "02:42:AC:12:00:04",
      "origin": "dynamic",
      "expiration-time": "2019-07-16T01:20:48.597Z"
    },
    {
      "ipv4-address": "172.18.0.6",
      "link-layer-address": "02:42:AC:12:00:06",
      "origin": "dynamic",
      "expiration-time": "2019-07-16T01:20:48.599Z"
    }
  ]
},
"ipv6": {
  "dhcp-client": true,
  "address": [
    {
      "ip-prefix": "2001:172:18::5/80",
      "origin": "dhcp",
      "status": "preferred"
    },
    {
      "ip-prefix": "fe80::42:acff:fe12:5/64",
      "origin": "link-layer",
      "status": "preferred"
    }
  ],
  "srl_nokia-interfaces-nbr:neighbor-discovery": {
    "dup-addr-detect": true,
    "reachable-time": 30,
    "stale-time": 14400
  }
}
}
}
},
"id": 0,
"jsonrpc": "2.0"
}

```

## Example 2: Multiple get request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "get",
  "params": {
    "commands": [
      {
        "path": "/interface[name=mgmt0]",
        "datastore": "state",
        "recursive": false
      },
      {
        "path": "/interface[name=ethernet-1/10]",
        "datastore": "state",
        "recursive": false
      }
    ]
  }
}
```

## Response (multiple get)

```
{
  "result": [
    {
      "name": "mgmt0",
      "admin-state": "enable",
      "mtu": 1514,
      "ifindex": 524304383,
      "oper-state": "up",
      "last-change": "2019-07-12T16:53:39.291Z"
    },
    {
      "name": "ethernet-1/10",
      "description": "dut2-dut4-2",
      "admin-state": "enable",
      "mtu": 9232,
      "ifindex": 180223,
      "oper-state": "up",
      "last-change": "2019-07-12T16:54:15.602Z"
    }
  ],
  "id": 0,
  "jsonrpc": "2.0"
}
```

## 5.5.2 JSON set example

The **set** method lets you set a configuration or run operational commands.

- Example 1: shows a multiple set method using update and replace.
- Example 2: shows a set method using **path-keywords**.
- Example 3: shows a set method using an action update.

### Example 1: Multiple set with update and replace request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "set",
  "params": {
    "commands": [
      {
        "action": "update",
        "path": "/interface[name=mgmt0]/description:my-description"
      },
      {
        "action": "replace",
        "path": "/interface[name=mgmt0]/subinterface[index=0]/description:my-subdescription"
      }
    ]
  }
}
```

### Response (multiple set)

```
{
  "result": {},
  "id": 0,
  "jsonrpc": "2.0"
}
```

### Example 2: Set method using path keywords request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "set",
  "params": {
    "commands": [
      {
        "action": "update",
        "path": "/interface[name={name}]/description:my-description",
        "path-keywords": {
          "name": "mgmt0"
        }
      }
    ]
  }
}
```

```
{
  "action": "replace",
  "path": "/interface[name={name}]/subinterface[index={index}]/description:my-subdescription",
  "path-keywords": {
    "name": "mgmt0",
    "index": "0"
  }
}
]
```

### Response (set using path-keywords)

```
{
  "result": {},
  "id": 0,
  "jsonrpc": "2.0"
}
```

### Example 3: Set using alternative update (specifying a value) request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "set",
  "params": {
    "commands": [
      {
        "action": "update",
        "path": "/interface[name=mgmt0]",
        "value": {
          "description": "my-description",
          "subinterface": {
            "index": "0",
            "description": "my-subdescription"
          }
        }
      }
    ]
  }
}
```

### Response (set using alternative update)

```
{
  "result": {},
  "id": 0,
  "jsonrpc": "2.0"
}
```

### 5.5.3 JSON delete example

The set method lets you utilize an action delete command to delete nodes or leafs within a configuration.

- Example 1: shows a multiple set delete request to delete the specified paths.

#### Example 1: multiple set method delete request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "set",
  "params": {
    "commands": [
      {
        "action": "delete",
        "path": "/interface[name=mgmt0]/description"
      },
      {
        "action": "delete",
        "path": "/interface[name=mgmt0]/subinterface[index=0]/description"
      }
    ]
  }
}
```

#### Response (multiple set delete)

```
{
  "result": {},
  "id": 0,
  "jsonrpc": "2.0"
}
```

### 5.5.4 JSON validate example

The validate method lets you verify that the system will accept a configuration transaction before applying it to the system. The 'delete', 'replace', and 'update' actions can be used with the validate method.

- Example 1: shows a validate request to delete a specified path.
- Example 2: shows a validate request to update and replace a specified path.

### Example 1: validate a delete request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "validate",
  "params": {
    "commands": [
      {
        "action": "delete",
        "path": "/interface[name=mgmt0]/description"
      }
    ],
    "datastore": "candidate"
  }
}
```

### Response (validate delete)

```
{
  "result": {},
  "id": 0,
  "jsonrpc": "2.0"
}
```

### Example 2: validate a update and replace request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "validate",
  "params": {
    "commands": [
      {
        "action": "update",
        "path": "/interface[name=mgmt0]/description:my-description"
      },
      {
        "action": "replace",
        "path": "/interface[name=mgmt0]/subinterface[index=0]
/description:my-subdescription"
      }
    ]
  }
}
```

### Response (validate update and delete)

```
{
  "result": {},
  "id": 0,
  "jsonrpc": "2.0"
}
```

## 5.5.5 JSON CLI example

The **cli** method lets you run CLI commands. While get and set methods are restricted to accessing data structures in the YANG models, the cli method can access commands that have been added to the system using python plug-ins.

- Example 1: shows an example of a JSON cli command request.
- Example 2: shows an example of a JSON cli command requesting the output format as text.

You can optionally define these output formats: json, text, or table. JSON is the default.

### Example 1: CLI method input request

```
{
  "jsonrpc": "2.0",
  "id": 0,
  "method": "cli",
  "params": {
    "commands": [
      "enter candidate",
      "info interface mgmt0"
    ]
  }
}
```

### Response (CLI input)

```
{
  "result": [
    {},
    {
      "interface": [
        {
          "name": "mgmt0",
          "description": "my-description",
          "admin-state": "enable",
          "subinterface": [
            {
              "index": 0,
              "description": "my-subdescription",
              "admin-state": "enable",
              "ipv4": {
                "dhcp-client": true
              },
              "ipv6": {
                "dhcp-client": true
              }
            }
          ]
        }
      ]
    }
  ]
}
```

```
    ],  
    "id": 0,  
    "jsonrpc": "2.0"  
  }  
}
```

### Example 2: CLI method input request with output formatting

```
{  
  "jsonrpc": "2.0",  
  "id": 0,  
  "method": "cli",  
  "params": {  
    "commands": [  
      "enter candidate",  
      "info interface mgmt0"  
    ],  
    "output-format": "text"  
  }  
}
```

### Response (CLI with output formatting)

```
{  
  "result": [  
    "",  
    "    interface mgmt0 {\n      description my-description\nadmin-state enable\n      subinterface 0 {\ndescription my-subdescription\nadmin-state enable\n      ipv4 {\ndhcp-client true\n      }\n      ipv6 {\ndhcp-client true\n      }\n    }\n  ],  
  "id": 0,  
  "jsonrpc": "2.0"  
}
```

## 6 CLI plug-ins

The SR Linux CLI is a python application that can load dynamic libraries from other applications. This flexibility allows users to create their own show commands, and also allows each application to own its own show command tree.

In addition to custom show commands, Nokia provides the following pre-defined plug-ins:

- [General and operational commands](#)
- [Show reports](#)

For details on how to create custom show routines, and define the classes and utility functions used to create them, see the *SR Linux CLI Plug-in Guide*.



**Note:** Command outputs shown in this guide are examples only; actual output may differ depending on supported functionality and user configuration.

### 6.1 General and operational commands

[Table 18](#) defines general and operational commands that can be entered at any point in the CLI.

**Table 18** General and operational commands

| Command             | Description                                                                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <TAB>               | Auto-completes a command                                                                                                                                                                                                            |
| /                   | Moves to the root; can also be used to reset the context to the root for a specific command (for example: <b>info from state /</b> )                                                                                                |
| ?                   | Displays context-based help                                                                                                                                                                                                         |
| back                | Returns to the context before executing the last command                                                                                                                                                                            |
| baseline <argument> | Arguments: <ul style="list-style-type: none"> <li>• check - check baseline update for current candidates</li> <li>• diff - show baseline configuration changes</li> <li>• update - update baseline for current candidate</li> </ul> |
| bash                | Opens a bash session                                                                                                                                                                                                                |

**Table 18 (Continued) General and operational commands**

| Command                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bash <command>                        | Executes a command without entering a bash session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| cls                                   | Clears the screen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| diff [flat]                           | Compares the current candidate against the running configuration. Specifying <code>flat</code> provides a copy/paste format showing inserts and deletes. Global arguments include: <ul style="list-style-type: none"> <li>• <code>baseline</code> - configuration candidate baseline</li> <li>• <code>candidate</code> - configuration candidate</li> <li>• <code>checkpoint</code> - configuration checkpoint</li> <li>• <code>factory</code> - factory configuration</li> <li>• <code>file</code> - file with configuration stored in .json format</li> <li>• <code>from</code> - source datastore to compare with</li> <li>• <code>rescue</code> - rescue configuration</li> <li>• <code>running</code> - running datastore</li> <li>• <code>startup</code> - startup configuration</li> </ul> If arguments are not used, this command must be used in candidate mode. |
| echo                                  | Echoes text back to a session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| enter                                 | Switches to a different mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| enter candidate                       | Enters the shared candidate mode. Shared candidate mode is the default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| enter candidate exclusive             | Enters the exclusive candidate mode<br>To switch between shared and exclusive, you must switch to a different datastore (for example. running).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| enter candidate exclusive name <name> | Enters the exclusive mode for a named candidate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| enter candidate name <name>           | Enters the shared candidate mode for a named shared candidate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| enter candidate private               | Enters the candidate private mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| enter candidate private name <name>   | Enters the candidate private mode for a named candidate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| enter running                         | Enters the running mode (default)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| enter show                            | Enters the show mode (used with show CLI plug-ins)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| enter state                           | Enters the state mode (all configuration and operational states)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| environment                           | Configures and displays environment variables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Table 18 (Continued) General and operational commands**

| Command                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| environment alias                       | Creates or overwrites an alias                                                                                                                                                                                                                                                                                                                                                                                                                            |
| environment bottom-toolbar              | Changes the text displayed in the bottom toolbar                                                                                                                                                                                                                                                                                                                                                                                                          |
| environment cli-engine type             | Sets cli engine type for interactive logins: <ul style="list-style-type: none"> <li>• basic</li> <li>• advanced</li> </ul>                                                                                                                                                                                                                                                                                                                                |
| environment complete-on-space           | Triggers auto-completion when a space is typed (default is to explicitly require a <TAB>)                                                                                                                                                                                                                                                                                                                                                                 |
| environment delete                      | Resets and removes environment settings                                                                                                                                                                                                                                                                                                                                                                                                                   |
| environment key-completer-limit <limit> | Number of keys limited in auto-completion                                                                                                                                                                                                                                                                                                                                                                                                                 |
| environment load                        | Loads the environment settings from a file: <ul style="list-style-type: none"> <li>• file (path of the configuration file)</li> <li>• home (use ~/.srlinuxrc configuration file)</li> </ul>                                                                                                                                                                                                                                                               |
| environment output-format               | Allows the default output format to change between text and JSON                                                                                                                                                                                                                                                                                                                                                                                          |
| environment prompt                      | Changes the prompt displayed before every input line                                                                                                                                                                                                                                                                                                                                                                                                      |
| environment save                        | Saves the current environment                                                                                                                                                                                                                                                                                                                                                                                                                             |
| environment save home                   | Saves the current environment to the user home directory                                                                                                                                                                                                                                                                                                                                                                                                  |
| environment save file <file>            | Saves the current environment to a specific file                                                                                                                                                                                                                                                                                                                                                                                                          |
| environment show                        | Shows the currently active environment settings                                                                                                                                                                                                                                                                                                                                                                                                           |
| exit                                    | Exits to a previous context                                                                                                                                                                                                                                                                                                                                                                                                                               |
| exit to <ancestor>                      | Exits to a specific ancestor of the current context                                                                                                                                                                                                                                                                                                                                                                                                       |
| exit all                                | Exits to the root                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| filter                                  | Filters output for show and info commands.<br>Usage: filter [<node>] [depth <value>] [fields <value>] [keys-only] [non-zero] <ul style="list-style-type: none"> <li>• node - Positional node to filter on</li> <li>• depth - Filter out sub-nodes that are more than <i>n</i> levels deeper</li> <li>• fields - Fields to include</li> <li>• keys-only - Hide all fields</li> <li>• non-zero - Show only non-zero values (numeric values only)</li> </ul> |

**Table 18 (Continued) General and operational commands**

| Command                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| help                              | Displays mode-related help                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| history                           | Displays the command history list with line numbers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| history hot                       | Displays the top 5 most frequently used commands                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| history clear                     | Clears the history                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| info [ <i>path</i> ]              | Shows the value of all nodes and fields under the current context; optionally made more specific by a path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| info depth < <i>n</i> >           | Filters out sub-nodes that are deeper than the specified depth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| info detail                       | Shows also default values for unset fields                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| info flat                         | Shows each node or field as a single line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| info use-proto-json               | Shows the output as the JSON used for the protobuf messages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| info from < <i>mode</i> >         | Executes the info command from within the specified mode (either candidate, running, or state). The current context can be retrieved without a from argument.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| list                              | Show the keys of all nodes under the current context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| monitor [ <i>path</i> ]           | Monitors state changes within the current context; optionally made more specific by a path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| monitor recursive [ <i>path</i> ] | Includes children when monitoring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| monitor sample                    | Uses sampling instead of on-change monitoring (interval in seconds)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| ping                              | <p>Sends IPv4 ICMPv4 echo requests to network hosts.</p> <p>Usage: ping6 &lt;destination&gt; [-I &lt;value&gt;] [-M &lt;value&gt;] [-Q &lt;value&gt;] [-c &lt;value&gt;] [-i &lt;value&gt;] [-s &lt;value&gt;] [-t &lt;value&gt;] [network-instance &lt;value&gt;]</p> <ul style="list-style-type: none"> <li>• -I - Source interface/IP</li> <li>• -M - Path MTU discovery strategy</li> <li>• -M- pmtudisc_options - sets df-bit set</li> <li>• -Q - tos</li> <li>• -c - Number of ping requests</li> <li>• -i - Wait interval in seconds between each packet</li> <li>• -s - Packet size</li> <li>• -t - TTL (Time-To-Live)</li> </ul> |

**Table 18 (Continued) General and operational commands**

| Command       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ping6         | Sends IPv6 ICMPv6 echo requests to network hosts.<br>Usage: ping6 <destination> [-I <value>] [-M <value>] [-Q <value>] [-c <value>] [-i <value>] [-s <value>] [-t <value>] [network-instance <value>]<br><ul style="list-style-type: none"> <li>• -I - Source interface/IP</li> <li>• -M - Path MTU discovery strategy</li> <li>• -M- pmtudisc_options - sets df-bit set</li> <li>• -Q - tos</li> <li>• -c - Number of ping requests</li> <li>• -i - Wait interval seconds between each packet</li> <li>• -s - Packet size</li> <li>• -t - TTL (Time-To-Live)</li> </ul> |
| pwc           | Prints the current working context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| quit          | Closes the CLI session                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| save          | Saves the current datastore to the given file in JSON format.<br>Usage: save [detail] file <value> [from <running state>] [text]<br><ul style="list-style-type: none"> <li>• detail - Additionally saves the default values for unset fields</li> <li>• file - Output filename</li> <li>• from - Datastore used to retrieve data from</li> <li>• text - Use CLI (text) format instead of JSON</li> </ul>                                                                                                                                                                 |
| show          | Displays plug-in style show commands; see <a href="#">6.2 Show reports</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| source <file> | Executes a set of commands from a file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| tech-support  | Generates a technical support file.<br>Usage: tech-support [ignore-host-keys <value>] [max-time <value>] [network-instance <value>] [no-core] [scp-to <value>]<br><ul style="list-style-type: none"> <li>• ignore-host-keys - Skip security verification of remote ssh server key</li> <li>• max-time - Maximum time</li> <li>• network-instance - Network-instance for scp command</li> <li>• no-core - Skip inclusion of core files</li> <li>• scp-to - scp location, for example, user@server-ip:/tmp/</li> </ul>                                                     |
| tools         | Executes a tool command                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Table 18 (Continued) General and operational commands**

| Command                                                | Description                                                                                                                                                                                                                         |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| traceroute                                             | Prints the route packets trace to network host<br>Usage: <code>traceroute &lt;destination&gt;</code>                                                                                                                                |
| traceroute6                                            | Prints the route IPv6 packets trace to network host<br>Usage: <code>traceroute6 &lt;destination&gt;</code>                                                                                                                          |
| tcptraceroute                                          | tcptraceroute compatible wrapper for traceroute<br>Usage: <code>tcptraceroute &lt;destination&gt;</code>                                                                                                                            |
| tree <i>[path]</i>                                     | Shows the tree structure in the current context; optionally made more specific with a path                                                                                                                                          |
| tree flat                                              | Shows each structure on a single line                                                                                                                                                                                               |
| tree from <i>[mode]</i>                                | Retrieves the tree from the current context in another mode                                                                                                                                                                         |
| watch                                                  | Execute a program periodically                                                                                                                                                                                                      |
| <b>Within candidate mode only, the following apply</b> |                                                                                                                                                                                                                                     |
| !!                                                     | Appends a line to the annotation of the current node                                                                                                                                                                                |
| !!!                                                    | Replaces the annotation of the current node                                                                                                                                                                                         |
| commit now                                             | Applies the changes, exits candidate mode, and enters running mode                                                                                                                                                                  |
| commit stay                                            | Applies the changes and then remains in candidate mode.<br>Permitted additional arguments: <code>commit stay [save] [comment] [confirmed]</code>                                                                                    |
| commit save                                            | Applies the changes and then remains in candidate mode.<br>Permitted additional arguments: <code>commit [stay] [checkpoint] save [confirmed] [comment]</code>                                                                       |
| commit checkpoint                                      | Causes an automatic checkpoint after the commit succeeds.<br>Permitted additional arguments: <code>commit [stay] [now] checkpoint [save] [confirmed]</code>                                                                         |
| commit validate                                        | Verifies that a propose configuration change passes a management server validation                                                                                                                                                  |
| commit comment<br><comment.>                           | Used with other arguments (except <b>validate</b> ) to add a user comment where comment is a quoted string, 1-255 characters.<br>Permitted additional arguments: <code>commit [stay] [save] [checkpoint] [confirmed] comment</code> |

**Table 18 (Continued) General and operational commands**

| Command                              | Description                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| commit confirmed                     | Applies the changes, but requires an explicit confirmation to become permanent.                                                                                                                             |
| commit confirmed [timeout <1-86400>] | The timeout period default is 600 seconds (10 mins.), or can be provisioned with a value of 1-86400 sec.). The timeout argument cannot be used with the accept or reject parameter.                         |
| commit confirmed [accept reject]     | Prior to the timer expiring, the accept argument explicitly confirms and applies the changes. With no timer running, the reject argument explicitly rejects the changes.                                    |
| annotate                             | Sets the annotation of the current node                                                                                                                                                                     |
| discard [now stay]                   | Discards all uncommitted changes; requires either a <b>now</b> or <b>stay</b> option (to stop unintended commit). When <b>stay</b> is used, the mode remains in candidate mode (opening a new transaction). |

---

## 6.2 Show reports

Show reports are python plug-ins used to create custom output. Nokia provides a set of pre-defined show reports. These pre-defined reports can be used as examples for how to create additional custom reports. See Chapter [7 Pre-defined show reports](#), for a complete list.

For more information about creating custom show reports, refer to the *SR Linux CLI Plug-In Guide*.

## 7 Pre-defined show reports

The following sections provide the commands and syntax for pre-defined python plug-ins. These commands output reports for the following areas: ACL, ARPND, interfaces, MPLS, network-instances, platform, systems, and versions.

Users can also create their own customer reports using CLI Plug-ins. For more information, refer to the *SR Linux CLI Plug-In Guide*.

### 7.1 ACL show reports

```
show
  — acl
    — capture-filter <filter name>
      — ipv4-filter <filter name>
        — entry <value>
      — ipv6-filter <filter name>
        — entry <value>
    — cpm-filter <filter name>
      — ipv4-filter <filter name>
        — entry <value>
      — ipv6-filter <filter name>
        — entry <entry number>
    — ipv4-filter <filter name>
      — entry <value>
      — subinterface <subinterface value>
    — ipv6-filter <filter name>
      — entry <value>
      — subinterface <subinterface value>
    — summary
```

#### 7.1.1 ACL descriptions

show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

## acl

|             |                                 |
|-------------|---------------------------------|
| Context     | <a href="#">show acl</a>        |
| Tree        | <a href="#">acl</a>             |
| Description | Container for ACL show reports. |

## capture-filter

|              |                                                                                |
|--------------|--------------------------------------------------------------------------------|
| Context      | <a href="#">show acl capture-filter</a> <i>filter type</i>                     |
| Tree         | <a href="#">capture-filter</a>                                                 |
| Description  | Show a list of ACL capture filters. Specify either ipv4-filter or ipv6-filter. |
| Configurable | False                                                                          |

## ipv4-filter

|             |                                                     |
|-------------|-----------------------------------------------------|
| Context     | <a href="#">show acl capture-filter ipv4-filter</a> |
| Tree        | <a href="#">ipv4-filter</a>                         |
| Description | Show a list of ACL capture IPv4 filters.            |

## entry

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Context     | <a href="#">show acl capture-filter ipv4-filter entry</a> <i>number</i> |
| Tree        | <a href="#">entry</a>                                                   |
| Description | Show a specific ACL capture IPv4 filter.                                |

## ipv6-filter

|             |                                                     |
|-------------|-----------------------------------------------------|
| Context     | <a href="#">show acl capture-filter ipv6-filter</a> |
| Tree        | <a href="#">ipv6-filter</a>                         |
| Description | Show a list of ACL capture IPv6 filters.            |

## entry

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Context     | <a href="#">show acl capture-filter ipv6-filter entry</a> <i>number</i> |
| Tree        | <a href="#">entry</a>                                                   |
| Description | Show a specific ACL capture IPv6 filter.                                |

## cpm-filter

|             |                                                                            |
|-------------|----------------------------------------------------------------------------|
| Context     | <a href="#">show acl cpm-filter</a> <i>filter type</i>                     |
| Tree        | <a href="#">cpm-filter</a>                                                 |
| Description | Show a list of ACL CPM filters. Specify either ipv4-filter or ipv6-filter. |

## ipv4-filter

|             |                                                 |
|-------------|-------------------------------------------------|
| Context     | <a href="#">show acl cpm-filter ipv4-filter</a> |
| Tree        | <a href="#">ipv4-filter</a>                     |
| Description | Show a list of ACL CPM IPv4 filters.            |

## entry

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| Context     | <a href="#">show acl cpm-filter ipv4-filter entry</a> <i>number</i> |
| Tree        | <a href="#">entry</a>                                               |
| Description | Show a specific ACL CPM IPv4 filter.                                |

## ipv6-filter

|             |                                                 |
|-------------|-------------------------------------------------|
| Context     | <a href="#">show acl cpm-filter ipv6-filter</a> |
| Tree        | <a href="#">ipv6-filter</a>                     |
| Description | Show a list of ACL CPM IPv6 filters.            |

## entry

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| Context     | <a href="#">show acl cpm-filter ipv6-filter entry</a> <i>number</i> |
| Tree        | <a href="#">entry</a>                                               |
| Description | Show a specific ACL CPM IPv6 filter.                                |

## ipv4-filter

|             |                                         |
|-------------|-----------------------------------------|
| Context     | <code>show acl ipv4-filter name</code>  |
| Tree        | <code>ipv4-filter</code>                |
| Description | Show ACL IPv4 filter policies.<br>False |

## entry

|             |                                                     |
|-------------|-----------------------------------------------------|
| Context     | <code>show acl ipv4-filter name entry number</code> |
| Tree        | <code>entry</code>                                  |
| Description | Show ACL IPv4 filter policies by entry ID.          |

## subinterface

|             |                                                            |
|-------------|------------------------------------------------------------|
| Context     | <code>show acl ipv4-filter name subinterface string</code> |
| Tree        | <code>subinterface</code>                                  |
| Description | Show ACL IPv4 filter policies by subinterface.             |

## ipv6-filter

|              |                                        |
|--------------|----------------------------------------|
| Context      | <code>show acl ipv6-filter name</code> |
| Tree         | <code>ipv6-filter</code>               |
| Description  | Show ACL IPv6 filter policies.         |
| Configurable | False                                  |

## entry

|             |                                                     |
|-------------|-----------------------------------------------------|
| Context     | <code>show acl ipv6-filter name entry number</code> |
| Tree        | <code>entry</code>                                  |
| Description | Show ACL IPv6 filter policies by entry ID.          |

## subinterface

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| Context     | <a href="#">show acl ipv6-filter</a> <i>name</i> <a href="#">subinterface</a> <i>string</i> |
| Tree        | <a href="#">subinterface</a>                                                                |
| Description | Show ACL IPv6 filter policies by subinterface.                                              |

## summary

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Context     | <a href="#">show acl summary</a>                                          |
| Tree        | <a href="#">summary</a>                                                   |
| Description | Show a summarized list of all ACL filters (capture, CPM, IPv4, and IPv6). |

## 7.2 ARPND show reports

```
show
  — arpnd
    — neighbors
      — interface <interface name>
    — arp-entries
      — interface <interface name>
```

### 7.2.1 ARPND descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

#### arpnd

|             |                                   |
|-------------|-----------------------------------|
| Context     | <a href="#">show arpnd</a>        |
| Tree        | <a href="#">arpnd</a>             |
| Description | Container for ARPND show reports. |

#### neighbors

|             |                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show arpnd neighbors</a>                                                                    |
| Tree        | <a href="#">neighbors</a>                                                                               |
| Description | Show a report of all ARPND neighbors including associated interfaces, subinterfaces, origin, and state. |

#### interface

|             |                                                                      |
|-------------|----------------------------------------------------------------------|
| Context     | <a href="#">show arpnd neighbors interface</a> <i>interface name</i> |
| Tree        | <a href="#">interface</a>                                            |
| Description | Show a ARPND neighbor report for a specified interface name.         |

## arp-entries

|             |                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show arpnd arp-entries</a>                                                                 |
| Tree        | <a href="#">arp-entries</a>                                                                            |
| Description | Show a report of all ARPND entries including associated interfaces, subinterfaces, origin, and expiry. |

## interface

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| Context     | <a href="#">show arpnd arp-entries interface</a> <i>interface name</i> |
| Tree        | <a href="#">interface</a>                                              |
| Description | Show a ARPND entries report for a specified interface name.            |

---

## 7.3 Interface show reports

```
show
  — interface <interface port name>
    — brief
    — all
    — detail
    — queue-details
```

### 7.3.1 Interface descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <b>show</b>                 |
| Tree        | <b>show</b>                 |
| Description | Container for show reports. |

#### interface

|             |                                                                                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show interface</b> <i>interface port name</i>                                                                                                                                                                                                                                                                                 |
| Tree        | <b>interface</b>                                                                                                                                                                                                                                                                                                                 |
| Description | Container for Interface show reports. Can be used to show a report of all interfaces and subinterfaces with an operational state of up, or for a specified interface if an optional name is entered. Interfaces that are operationally down are not displayed; use the <b>all</b> option to display both up and down interfaces. |

#### brief

|             |                                                                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show interface</b> <i>interface port name</i> <b>brief</b>                                                                                                                                                          |
| Tree        | <b>brief</b>                                                                                                                                                                                                           |
| Description | Show a report that provides a summarized view of all interfaces and subinterfaces, or for a specified interface if an optional name is entered. Displays the administrative state, operational state, speed, and type. |

## all

|             |                                                                                                                                                                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show interface</b> <i>interface port name</i> <b>all</b>                                                                                                                                                                                                                                                                             |
| Tree        | <b>all</b>                                                                                                                                                                                                                                                                                                                              |
| Description | Show a report that displays all up and down interfaces and subinterfaces, or for a specified interface if an optional name is entered. For interfaces with an operational state of up, IP addresses are displayed in addition to speed and type. For interfaces with an operational state of down, a reason for this state is provided. |

## detail

|             |                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show interface</b> <i>interface port name</i> <b>detail</b>                                                                                                                                                                                                                                                            |
| Tree        | <b>detail</b>                                                                                                                                                                                                                                                                                                             |
| Description | Show a detail report for all up and down interfaces and subinterfaces, or for a specified interface if an optional name is entered. In addition to operational state, last changed, flow control and mac address details, this report includes queue parameters, statistics, and transceiver/transceiver channel details. |

## queue-details

|             |                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show interface</b> <i>interface port name</i> <b>queue-details</b>                                                                    |
| Tree        | <b>queue-details</b>                                                                                                                     |
| Description | Show an egress queues and VOQs report for all interfaces and subinterfaces, or for a specified interface if an optional name is entered. |

## 7.4 LAG show reports

```
show
  — lag <lag instance>
    — brief
    — detail
    — lacp-state
    — lacp-statistics
    — member-statistics
    — queue-details
```

### 7.4.1 LAG descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

#### lag

|             |                                                                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show lag</a> <i>lag instance</i>                                                                                                                                                                  |
| Tree        | <a href="#">lag</a>                                                                                                                                                                                           |
| Description | Container for lag show reports. Can be used to show a report of all LAGs, or for a specified LAG instance if an optional ID is entered. This includes a summary report of all operationally up and down LAGs. |

#### brief

|             |                                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show lag</a> <i>lag instance</i> <a href="#">brief</a>                                                                                                                                               |
| Tree        | <a href="#">brief</a>                                                                                                                                                                                            |
| Description | Show a report that provides a summarized view of all LAGs, or for a specified LAG if an optional ID is entered. Displays the administrative state, operational state, aggregate speed, and min and active links. |

---

## detail

|             |                                                                                                                                                                                                                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show lag lag instance detail</b>                                                                                                                                                                                                                                                               |
| Tree        | <b>detail</b>                                                                                                                                                                                                                                                                                     |
| Description | Show a detail report for all LAGs, or for a specified LAG if an optional ID is entered. This report includes the operational status of the LAG, list of member links and their operational status, aggregated queue statistics, aggregated traffic/error statistics, and aggregated traffic rate. |

## lACP-state

|             |                                                                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show lag lag instance lACP-state</b>                                                                                                                                                                            |
| Tree        | <b>lACP-state</b>                                                                                                                                                                                                  |
| Description | Show a LAG report that details all LAGs or a specific LAG it is enabled. Report also includes the LACP configuration mode and per member link statistics state details (operation state, activity, timeout, etc.). |

## lACP-statistics

|             |                                                                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show lag lag instance lACP-statistics</b>                                                                                                                                                              |
| Tree        | <b>lACP-statistics</b>                                                                                                                                                                                    |
| Description | Show a LAG report that details all LAGs or specific LAGs statistics. Report can include per member link statistics such as LACP in-pkts, out-pkts, rx-errors, tx-errors, unknown errors, and LACP errors. |

## member-statistics

|             |                                                        |
|-------------|--------------------------------------------------------|
| Context     | <b>show lag lag instance member-statistics</b>         |
| Tree        | <b>member-statistics</b>                               |
| Description | Show a LAG report that displays all member-statistics. |

## queue-details

|             |                                                                                       |
|-------------|---------------------------------------------------------------------------------------|
| Context     | <b>show lag lag instance queue-details</b>                                            |
| Tree        | <b>queue-details</b>                                                                  |
| Description | Show a LAG report that displays aggregated queue statistics for all LAG link members. |

---

## 7.5 MPLS show reports

```
show
  — mpls-aft
    — network-instance <instance name>
```

### 7.5.1 MPLS descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

#### mpls-aft

|             |                                                                                      |
|-------------|--------------------------------------------------------------------------------------|
| Context     | <a href="#">show mpls-aft</a>                                                        |
| Tree        | <a href="#">mpls-aft</a>                                                             |
| Description | Container for MPLS reports. Can be used to show all MPLS abstract forwarding tables. |

#### network-instance

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| Context     | <a href="#">show mpls-aft network-instance</a> <i>network-instance name</i> |
| Tree        | <a href="#">network-instance</a>                                            |
| Description | Show an MPLS abstract forwarding table for a specified network instance.    |

## 7.6 Network-instance show reports

```
show
  — network-instance <name>
    — bridge-table
      — mac-duplication duplication-entries
    — mac-table
      — all
      — mac <address>
      — summary
  — interfaces <interface name>
  — protocols
    — bgp
      — neighbor [<IP-address>]
        — advertised-routes
          — evpn
          — ipv4
          — ipv6
        — detail
        — maintenance
        — received-routes
          — evpn
          — ipv4
          — ipv6
      — routes [<IP family>]
        — evpn
          — route-type <route type>
            — detail
            — summary
          — ipv4
            — prefix <IP prefix>
            — summary
          — ipv6
            — prefix <IP prefix>
            — summary
        — summary
    — bgp-evpn
    — bgp-instance
    — bgp-vpn
    — bgp-instances
  — isis
    — adjacency <interface_name>
      — neighbor-system-id <ID>
        — adjacency-level <value>
        — detail
    — database
      — lsp-id <ID>
      — detail
    — interface <name>
      — detail
    — hostnames
      — detail
```

- summary
- ospf
  - area
    - detail
  - database
    - type
  - instance <name>
  - interface
    - detail
  - neighbor
    - detail
  - statistics
  - status
- route-table
  - all
  - summary
  - ipv4-unicast
    - prefix <ID> detail
    - summary
  - ipv6-unicast
    - prefix <ID> detail
    - summary
  - next-hop <index>
- summary
- tunnel-table
  - all
  - ipv4 <ip address> type
  - ipv6 <ip address> type
- vxlan-interface <name>

## 7.6.1 Network-instance descriptions

### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

### network-instance

|             |                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance</a> <i>name</i>                                                                             |
| Tree        | <a href="#">network-instance</a>                                                                                              |
| Description | Show network-instance for specified name or additional command (bridge-table, interfaces, protocols, route-table, or summary) |

## bridge-table

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>bridge-table</b>                      |
| Tree        | <b>bridge-table</b>                                                               |
| Description | Show MAC bridge tables. Specify mac-duplication duplication-entries or mac-table. |

## mac-duplication duplication-entries

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>bridge-table mac-duplication duplication-entries</b> |
| Tree        | <b>mac-duplication duplication-entries</b>                                                       |
| Description | Show all MAC learned entries.                                                                    |

## mac-table

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>bridge-table mac-table</b> |
| Tree        | <b>mac-table</b>                                                       |
| Description | Show a MAC table report. Specify option of all, mac, or summary.       |

## all

|             |                                                                            |
|-------------|----------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>bridge-table mac-table all</b> |
| Tree        | <b>all</b>                                                                 |
| Description | Show all MAC table entries                                                 |

## mac

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>bridge-table mac-table mac</b> <i>address</i> |
| Tree        | <b>mac</b>                                                                                |
| Description | Show MAC table entry for specific address.                                                |

## summary

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>bridge-table mac-table summary</b> |
| Tree        | <b>summary</b>                                                                 |
| Description | Show a MAC table summary report.                                               |

---

## interfaces

|             |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>interfaces</b> <i>interface name</i> |
| Tree        | <b>interfaces</b>                                                                |
| Description | Show all network-instance interfaces.                                            |

## protocols

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols</b>           |
| Tree        | <b>protocols</b>                                                    |
| Description | Show network-instance protocol details. Specify bgp, isis, or ospf. |

## bgp

|             |                                                               |
|-------------|---------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp</b> |
| Tree        | <b>bgp</b>                                                    |
| Description | Show bgp status report. Specify neighbor, routes, or summary. |

## neighbor

|             |                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i>      |
| Tree        | <b>neighbor</b>                                                                               |
| Description | Show a bgp neighbor report. Specify a peer address to see a report for the specified address. |

## advertised-routes

|             |                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>advertised-routes</b>       |
| Tree        | <b>advertised-routes</b>                                                                                                |
| Description | Show a bgp neighbor advertised routes report. The network-instance name and neighbor peer IP address must be specified. |

## evpn

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>advertised-routes evpn</b>           |
| Tree        | <b>evpn</b>                                                                                                                      |
| Description | Show a bgp neighbor advertised routes report for EVPN. The network-instance name and neighbor peer IP address must be specified. |

## ipv4

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>advertised-routes ipv4</b>           |
| Tree        | <b>ipv4</b>                                                                                                                      |
| Description | Show a bgp neighbor advertised routes report for IPv4. The network-instance name and neighbor peer IP address must be specified. |

## ipv6

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>advertised-routes ipv6</b>           |
| Tree        | <b>ipv6</b>                                                                                                                      |
| Description | Show a bgp neighbor advertised routes report for IPv6. The network-instance name and neighbor peer IP address must be specified. |

## maintenance

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>maintenance</b> |
| Tree        | <b>maintenance</b>                                                                                          |
| Description | Show a bgp neighbor report that shows the maintenance mode and group.                                       |

## received-routes

|             |                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>received-routes</b>   |
| Tree        | <b>received-routes</b>                                                                                            |
| Description | Show a bgp neighbor received routes report. network-instance name and neighbor peer IP address must be specified. |

## evpn

|             |                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>received-routes evpn</b>           |
| Tree        | <b>evpn</b>                                                                                                                    |
| Description | Show a bgp neighbor received routes report for EVPN. The network-instance name and neighbor peer IP address must be specified. |

## ipv4

|             |                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>received-routes ipv4</b>           |
| Tree        | <b>ipv4</b>                                                                                                                    |
| Description | Show a bgp neighbor received routes report for IPv4. The network-instance name and neighbor peer IP address must be specified. |

## ipv6

|             |                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>received-routes ipv6</b>           |
| Tree        | <b>ipv6</b>                                                                                                                    |
| Description | Show a bgp neighbor received routes report for IPv6. The network-instance name and neighbor peer IP address must be specified. |

## detail

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp neighbor</b> <i>IP address</i> <b>detail</b>     |
| Tree        | <b>detail</b>                                                                                              |
| Description | Show a bgp neighbor detailed report. network-instance name and neighbor peer IP address must be specified. |

## routes

|             |                                                                      |
|-------------|----------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes</b> |
| Tree        | <b>routes</b>                                                        |
| Description | Show bgp route report for an IP family (EVPN, IPv4 or IPv6).         |

## evpn

|             |                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes evpn</b>                                                                |
| Tree        | <b>evpn</b>                                                                                                                              |
| Description | Show a bgp EVPN route report. Additional parameters can define a report for a specific route type, and a detailed versus summary report. |

## route-type

|             |                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes evpn route-type</b> <i>route type</i>                                 |
| Tree        | <b>route-type</b>                                                                                                                      |
| Description | Show a bgp EVPN route report for a specific route type. Route type can be numeric or of type esi, mac-address, and originating-router. |

## detail

|             |                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes evpn route-type</b> <i>route type</i> <b>detail</b> |
| Tree        | <b>detail</b>                                                                                                        |
| Description | Show a detailed bgp EVPN route report for a specific route type.                                                     |

## summary

|             |                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes evpn route-type</b> <i>route type</i> <b>summary</b> |
| Tree        | <b>summary</b>                                                                                                        |
| Description | Show a summarized bgp EVPN route report for a specific route type.                                                    |

## ipv4

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes ipv4</b> |
| Tree        | <b>ipv4</b>                                                               |
| Description | Show a bgp IPv4 route report. Specify an IP prefix or summary.            |

## prefix

|             |                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes ipv4 prefix</b> <i>IP prefix</i> |
| Tree        | <b>prefix</b>                                                                                     |
| Description | Show a bgp IPv4 route report for a specified IP prefix.                                           |

## summary

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes ipv4 summary</b> |
| Tree        | <b>summary</b>                                                                    |
| Description | Show a bgp IPv4 route summary report.                                             |

## ipv6

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes ipv6</b> |
| Tree        | <b>ipv6</b>                                                               |
| Description | Show a bgp IPv6 route report. Specify an IP prefix or summary.            |

## prefix

|             |                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes ipv6 prefix</b> <i>IP prefix</i> |
| Tree        | <b>prefix</b>                                                                                     |
| Description | Show a bgp IPv6 route report for a specified IP prefix.                                           |

## summary

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp routes ipv6 summary</b> |
| Tree        | <b>summary</b>                                                                    |
| Description | Show a bgp IPv6 route summary report.                                             |

## summary

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp summary</b> |
| Tree        | <b>summary</b>                                                        |
| Description | Show a bgp summary report.                                            |

## bgp-evpn

|             |                                                                    |
|-------------|--------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp-evpn</b> |
| Tree        | <b>bgp-evpn</b>                                                    |
| Description | Show a bgp-evpn status report.                                     |

## bgp-instance

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp-evpn bgp-instance</b> <i>instance</i> |
| Tree        | <b>bgp-instance</b>                                                                             |
| Description | Show a bgp-evpn status report for a specific bgp instance.                                      |

## bgp-vpn

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp-vpn</b> |
| Tree        | <b>bgp-vpn</b>                                                    |
| Description | Show a bgp-vpn status report.                                     |

## bgp-instance

|             |                                                                                                |
|-------------|------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols bgp-vpn bgp-instance</b> <i>instance</i> |
| Tree        | <b>bgp-instance</b>                                                                            |
| Description | Show a bgp-vpn status report for a specific bgp instance.                                      |

## isis

|             |                                                                                         |
|-------------|-----------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis</b>                          |
| Tree        | <b>isis</b>                                                                             |
| Description | Show ISIS status report. Specify adjacency, database, hostnames, interface, or summary. |

## adjacency

|             |                                                                                                |
|-------------|------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis adjacency</b> <i>interface_name</i> |
| Tree        | <b>adjacency</b>                                                                               |
| Description | Show a list of ISIS adjacencies formed through this interface.                                 |

---

## neighbor-system-id

|             |                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis adjacency</b> <i>interface_name</i><br><b>neighbor-system-id</b> <i>ID</i> |
| Tree        | <b>neighbor-system-id</b>                                                                                                             |
| Description | Show a list of ISIS adjacencies for a specified system ID of a neighbor router.                                                       |

## adjacency-level

|             |                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis adjacency</b> <i>interface_name</i><br><b>neighbor-system-id</b> <i>ID</i> <b>adjacency-level</b> <i>value</i> |
| Tree        | <b>adjacency-level</b>                                                                                                                                                    |
| Description | Show a list of ISIS adjacencies for a specified adjacency level.                                                                                                          |

## detail

|             |                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis adjacency</b> <i>interface_name</i><br><b>neighbor-system-id</b> <i>ID</i> <b>adjacency-level</b> <i>value</i> <b>detail</b> |
| Tree        | <b>detail</b>                                                                                                                                                                           |
| Description | Show a detailed ISIS adjacency report.                                                                                                                                                  |

## database

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis database</b> |
| Tree        | <b>database</b>                                                         |
| Description | Show an ISIS database report.                                           |

## lsp-id

|             |                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis database lsp-id</b> <i>ID</i>                                                            |
| Tree        | <b>lsp-id</b>                                                                                                                                       |
| Description | Show an ISIS database report for a specified lsp-id (ID format: 6 octets of adjacency system-id followed by 1 octet Lan-ID and 1 octet LSP Number). |

## detail

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis database detail</b> |
| Tree        | <b>detail</b>                                                                  |
| Description | Show a detailed ISIS database report.                                          |

## interface

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis interface</b> <i>name</i>      |
| Tree        | <b>interface</b>                                                                          |
| Description | Show an ISIS interface report for all interfaces or optionally specify an interface name. |

## detail

|             |                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis interface</b> <i>name</i> <b>detail</b> |
| Tree        | <b>detail</b>                                                                                      |
| Description | Show a detailed ISIS interface report for all interfaces or optionally specify an interface name.  |

## hostnames

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis hostnames</b> |
| Tree        | <b>hostnames</b>                                                         |
| Description | Show an ISIS hostname report.                                            |

## detail

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis hostnames detail</b> |
| Tree        | <b>detail</b>                                                                   |
| Description | Show a detailed ISIS hostname report.                                           |

## summary

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols isis summary</b> |
| Tree        | <b>summary</b>                                                         |
| Description | Show a summarized ISIS report.                                         |

## ospf

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance <i>name</i> protocols ospf</a>       |
| Tree        | <a href="#">ospf</a>                                                   |
| Description | Show OSPF status report. Specify area, interface, neighbor, or status. |

## area

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| Context     | <a href="#">show network-instance <i>name</i> protocols ospf area</a> |
| Tree        | <a href="#">area</a>                                                  |
| Description | Show OSPF report for areas where the local system exists.             |

## detail

|             |                                                                              |
|-------------|------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance <i>name</i> protocols ospf area detail</a> |
| Tree        | <a href="#">detail</a>                                                       |
| Description | Show detailed OSPF report for areas where the local system exists.           |

## database

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance <i>name</i> protocols ospf database</a> |
| Tree        | <a href="#">database</a>                                                  |
| Description | Show OSPF database report.                                                |

## type

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance <i>name</i> protocols ospf database type</a>  |
| Tree        | <a href="#">type</a>                                                            |
| Description | Show OSPF database report for a specific database type. For example: router-lsa |

## instance

|             |                                                                                                                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols ospf instance</b> <i>instance name</i>                                                                                                                 |
| Tree        | <b>instance</b>                                                                                                                                                                                              |
| Description | Show OSPF report for a specific instance. After the instance name, a more detailed report can be generated by specifying area, database, interface, neighbor, statistics, or status after the instance name. |

## interface

|             |                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols ospf interface</b>                                         |
| Tree        | <b>interface</b>                                                                                                 |
| Description | Show OSPF report for OSPF interfaces that act as a connection between a router and one of its attached networks. |

## detail

|             |                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols ospf interface detail</b>                                           |
| Tree        | <b>detail</b>                                                                                                             |
| Description | Show detailed OSPF report for OSPF interfaces that act as a connection between a router and one of its attached networks. |

## neighbor

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols ospf neighbor</b> |
| Tree        | <b>neighbor</b>                                                         |
| Description | Show OSPF report for OSPF neighbors.                                    |

## detail

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>protocols ospf neighbor detail</b> |
| Tree        | <b>detail</b>                                                                  |
| Description | Show detailed OSPF report for OSPF neighbors.                                  |

---

## statistics

|             |                                                                    |
|-------------|--------------------------------------------------------------------|
| Context     | <b>show network-instance <i>name</i> protocols ospf statistics</b> |
| Tree        | <b>statistics</b>                                                  |
| Description | Show OSPF Rx, Tx, and packet statistics report.                    |

## status

|             |                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance <i>name</i> protocols ospf status</b>                                                             |
| Tree        | <b>status</b>                                                                                                              |
| Description | Show an overall status report for OSPF (router IDs, version, admin state, backbone and area border router settings, etc.). |

## route-table

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance <i>name</i> route-table</b>                                                        |
| Tree        | <b>route-table</b>                                                                                          |
| Description | Show a network-instance route table report. Specify all, ipv4-unicast, ipv6-unicast, next-hop, or, summary. |

## all

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance <i>name</i> route-table all</b>                                  |
| Tree        | <b>all</b>                                                                                |
| Description | Show a report of all routes that includes prefix, ID, active status and next-hop details. |

## summary

|             |                                                              |
|-------------|--------------------------------------------------------------|
| Context     | <b>show network-instance <i>name</i> route-table summary</b> |
| Tree        | <b>summary</b>                                               |
| Description | Show a summary report of active routes by name and protocol. |

## ipv4-unicast

|             |                                                                                            |
|-------------|--------------------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance</a> <i>name</i> <a href="#">route-table ipv4-unicast</a> |
| Tree        | <a href="#">ipv4-unicast</a>                                                               |
| Description | Show a IPv4 route table report. Specify a summary report or by prefix ID.                  |

## prefix <ID> detail

|             |                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance</a> <i>name</i> <a href="#">route-table ipv4-unicast</a> <a href="#">prefix &lt;ID&gt; detail</a>                |
| Tree        | <a href="#">prefix</a>                                                                                                                             |
| Description | Show a IPv4 route table report for a specified prefix. Report includes destination, ID, owner, active status and details when it was last changed. |

## summary

|             |                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance</a> <i>name</i> <a href="#">route-table ipv4-unicast</a> <a href="#">summary</a> |
| Tree        | <a href="#">summary</a>                                                                                            |
| Description | Show a IPv4 route table summary report that includes prefix, IDs, active status, and next-hop details.             |

## ipv6-unicast

|             |                                                                                            |
|-------------|--------------------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance</a> <i>name</i> <a href="#">route-table ipv6-unicast</a> |
| Tree        | <a href="#">ipv6-unicast</a>                                                               |
| Description | Show a IPv6 route table report. Specify a summary report or by prefix ID.                  |

## prefix <ID> detail

|             |                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show network-instance</a> <i>name</i> <a href="#">route-table ipv6-unicast</a> <a href="#">prefix &lt;ID&gt; detail</a>                |
| Tree        | <a href="#">prefix</a>                                                                                                                             |
| Description | Show a IPv6 route table report for a specified prefix. Report includes destination, ID, owner, active status and details when it was last changed. |

---

## summary

|             |                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>route-table ipv6-unicast summary</b>                       |
| Tree        | <b>summary</b>                                                                                         |
| Description | Show a IPv6 route table summary report that includes prefix, IDs, active status, and next-hop details. |

## next-hop

|             |                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>route-table next-hop</b> <i>index</i>                                              |
| Tree        | <b>next-hop</b>                                                                                                                |
| Description | Show a report of all next-hop route tables or optionally provide an index number to show a report for a specified route table. |

## summary

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>summary</b>                                          |
| Tree        | <b>summary</b>                                                                                   |
| Description | Show a summary report for all network-instances, or specify a network-name to refine the report. |

## tunnel-table

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>tunnel-table</b>                    |
| Tree        | <b>tunnel-table</b>                                                             |
| Description | Context for all tunnels in a tunnel table. Use with options all, IPv4, or IPv6. |

## all

|             |                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>tunnel-table all</b>                                                        |
| Tree        | <b>all</b>                                                                                                              |
| Description | Shows a network instance tunnel report that includes the prefix, type, ID, metric, preference, and last updated fields. |

## ipv4 *ip address* type

|             |                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>tunnel-table ipv4 &lt;ip address&gt; type</b>                                                                                                                                 |
| Tree        | <b>ipv4</b>                                                                                                                                                                                                               |
| Description | Show a network instance IPv4 tunnel report that includes the prefix, type, ID, metric, preference, and last updated fields. Report can be defined further by specifying the type (such as vxlan, ldp, and * (all types)). |

## ipv6 *ip address* type

|             |                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>tunnel-table ipv6 &lt;ip address&gt; type</b>                                                                                                                                 |
| Tree        | <b>ipv6</b>                                                                                                                                                                                                               |
| Description | Show a network instance IPv6 tunnel report that includes the prefix, type, ID, metric, preference, and last updated fields. Report can be defined further by specifying the type (such as vxlan, ldp, and * (all types)). |

## vxlan-interface

|             |                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show network-instance</b> <i>name</i> <b>vxlan-interface</b> <i>name</i>                                                                                                                        |
| Tree        | <b>vxlan-interface</b>                                                                                                                                                                             |
| Description | Show a network instance vxlan-interface report that defines the type, operation state and operation down reason for a specified interface or all interfaces (when the vxlan-interface name is ""). |

## 7.7 Platform show reports

```
show
  — platform
    — control <slot ID>
    — environment
    — fabric <slot number>
    — fan-tray <ID number>
    — linecard <slot number>
    — power-supply <ID number>
    — redundancy
    — resource-monitoring
```

### 7.7.1 Platform descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

#### platform

|             |                                         |
|-------------|-----------------------------------------|
| Context     | <a href="#">show platform</a>           |
| Tree        | <a href="#">platform</a>                |
| Description | Show a state report for all components. |

#### control

|             |                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform control</a> <i>slot ID</i>                                                                |
| Tree        | <a href="#">control</a>                                                                                             |
| Description | Show a report for the control module configuration and state. Shows all modules unless a specific slot is provided. |

## environment

|             |                                                                                        |
|-------------|----------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform environment</a>                                              |
| Tree        | <a href="#">environment</a>                                                            |
| Description | Show a report for the platform environment (state and temperature) for all components. |

## fabric

|             |                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform fabric slot ID</a>                                                                       |
| Tree        | <a href="#">fabric</a>                                                                                             |
| Description | Show a report for the fabric module configuration and state. Shows all modules unless a specific slot is provided. |

## fan-tray

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform fan-tray slot ID</a>                                                                |
| Tree        | <a href="#">fan-tray</a>                                                                                      |
| Description | Show a report for the fan tray configuration and state. Shows all modules unless a specific slot is provided. |

## linecard

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform linecard slot ID</a>                                                                |
| Tree        | <a href="#">linecard</a>                                                                                      |
| Description | Show a report for the linecard configuration and state. Shows all modules unless a specific slot is provided. |

## power-supply

|             |                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform power-supply ID</a>                                                                   |
| Tree        | <a href="#">power-supply</a>                                                                                    |
| Description | Show a report for the power-supply configuration and state. Shows all modules unless a specific ID is provided. |

## redundancy

|             |                                          |
|-------------|------------------------------------------|
| Context     | <a href="#">show platform redundancy</a> |
| Tree        | <a href="#">redundancy</a>               |
| Description | Show a platform redundancy report.       |

## resource-monitoring

|             |                                                                                         |
|-------------|-----------------------------------------------------------------------------------------|
| Context     | <a href="#">show platform resource-monitoring</a>                                       |
| Tree        | <a href="#">resource-monitoring</a>                                                     |
| Description | Show a resource-monitoring report for areas such as ACL, TCAM, IP MPLS forwarding, etc. |

## 7.8 System show reports

```
show
  - system
    - aaa authentication session <session ID number>
    - application <application name>
    - lldp neighbor
      - interface <interface name>
    - logging
      - buffer
        - messages
        - system
      - file
        - messages
    - network-instance ethernet-segments <name>
    - sflow status
```

### 7.8.1 System descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <b>show</b>                 |
| Tree        | <b>show</b>                 |
| Description | Container for show reports. |

#### system

|             |                                               |
|-------------|-----------------------------------------------|
| Context     | <b>show system</b>                            |
| Tree        | <b>system</b>                                 |
| Description | Container for system management show reports. |

#### aaa authentication session

|             |                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------|
| Context     | <b>show system aaa authentication session</b> <i>session ID</i>                                           |
| Tree        | <b>aaa authentication session</b>                                                                         |
| Description | Show a list of all active sessions in the system or for a specific session ID user if an ID is specified. |

## application

|             |                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show system application</b> <i>application name</i>                                                               |
| Tree        | <b>application</b>                                                                                                   |
| Description | Show a list of all applications and their status in the system or for a specific application if a name is specified. |

## lldp neighbor

|             |                                                                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show system lldp neighbor</b>                                                                                                     |
| Tree        | <b>lldp neighbor</b>                                                                                                                 |
| Description | Show a report of all LLDP neighbors that includes system name, chassis ID, first message, last update, port and related BGP details. |

## interface

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| Context     | <b>show system lldp neighbor interface</b> <i>interface name</i>            |
| Tree        | <b>interface</b>                                                            |
| Description | Show a report of a specific LLDP neighbors by specifying an interface name. |

## logging

|             |                                                        |
|-------------|--------------------------------------------------------|
| Context     | <b>show system logging</b>                             |
| Tree        | <b>logging</b>                                         |
| Description | Shows a list of system logs, type, and directory path. |

## buffer

|             |                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------|
| Context     | <b>show system logging buffer</b>                                                                    |
| Tree        | <b>buffer</b>                                                                                        |
| Description | Shows a report of type buffer log files maintained in memory (non-persistent across system reboots). |

---

## messages

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show system logging buffer messages</a>                                                        |
| Tree        | <a href="#">messages</a>                                                                                   |
| Description | Show a report of the messages associated with the system logging buffer (/var/log/srlinux/buffer/messages) |

## system

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show system logging buffer system</a>                                                             |
| Tree        | <a href="#">system</a>                                                                                        |
| Description | Show a report of the system related information in the system logging buffer (/var/log/srlinux/buffer/system) |

## file

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show system logging file</a>                                                        |
| Tree        | <a href="#">file</a>                                                                            |
| Description | Shows a report of log files that have been directed to a specific file (/var/log/srlinux/file). |

## messages

|             |                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show system logging file messages</a>                                                      |
| Tree        | <a href="#">messages</a>                                                                               |
| Description | Show a report of the messages associated with logs in a specific file (/var/log/srlinux/file/messages) |

## network-instance ethernet-segments

|             |                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show system network-instance ethernet-segments</a> <i>name</i>                                      |
| Tree        | <a href="#">network-instance ethernet-segments</a>                                                              |
| Description | Show a report that defines ethernet-segments details such as ESI, Alg, peers, and associated network instances. |

## sflow status

|             |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| Context     | <a href="#">show system sflow status</a>                                       |
| Tree        | <a href="#">sflow status</a>                                                   |
| Description | Show an sflow status report that includes the state, and sample size and rate. |

---

## 7.9 Tunnel show reports

```
show
  — tunnel
    — vxlan-tunnel
      — all
      — vtep <ip address>
```

### 7.9.1 Tunnel descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

#### tunnel

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show tunnel</a> |
| Tree        | <a href="#">tunnel</a>      |
| Description | Enter the tunnel context.   |

#### vxlan-tunnel

|             |                                                                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show tunnel vxlan-tunnel</a>                                                                                                                                                                  |
| Tree        | <a href="#">vxlan-tunnel</a>                                                                                                                                                                              |
| Description | Show a tunnel report specific to VXLAN tunnels. Report includes the VXLAN tunnel endpoint address, index and when last updated. Specify a specific VTEP ip address or 'all' to display all vxlan tunnels. |

## all

|             |                                                                                                                                      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show tunnel vxlan-tunnel all</a>                                                                                         |
| Tree        | <a href="#">all</a>                                                                                                                  |
| Description | Show a tunnel report specific to all VXLAN tunnels. Report includes the VXLAN tunnel endpoint address, index, and when last updated. |

## vtep

|             |                                                                                                                                    |
|-------------|------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show tunnel vxlan-tunnel vtep ip address</a>                                                                           |
| Tree        | <a href="#">vtep</a>                                                                                                               |
| Description | Show a tunnel report for a specific VXLAN tunnel. Report includes the VXLAN tunnel endpoint address, index, and when last updated. |

## 7.10 Tunnel-interface show reports

```
show
  — tunnel-interface <interface>
  — vxlan-interface <interface>
    — bridge-table
      — mac-table
      — multicast-destinations
        — destination | VNI
      — unicast-destinations
        — destination | VNI
    — brief
    — detail
```

### 7.10.1 Tunnel-interface descriptions

#### show

|             |                             |
|-------------|-----------------------------|
| Context     | <b>show</b>                 |
| Tree        | <b>show</b>                 |
| Description | Container for show reports. |

#### tunnel-interface

|             |                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i>                                                                                             |
| Tree        | <b>tunnel-interface</b>                                                                                                                   |
| Description | Enter the context for tunnel interface (reports for EVPN Layer-2). Specify a tunnel-interface name or an asterisk (*) for all interfaces. |

#### vxlan-interface

|             |                                                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i>                                                                                                   |
| Tree        | <b>vxlan-interface</b>                                                                                                                                                                  |
| Description | Show a vxlan interface report for EVPN-VXLAN for layer-2. Specify an vxlan-interface name or an asterisk (*) for all interfaces. Report options include brief, detail, or bridge-table. |

## bridge-table

|             |                                                                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>bridge-table</b>                                                                   |
| Tree        | <b>bridge-table</b>                                                                                                                                                         |
| Description | Show a bridge-table vxlan interface report for EVPN-VXLAN for layer-2. Options to use with bridge table include mac-table, multicast-destinations, and unicast-destination. |

## mac-table

|             |                                                                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>bridge-table mac-table</b>                              |
| Tree        | <b>mac-table</b>                                                                                                                                 |
| Description | Show a mac-table vxlan interface report for EVPN-VXLAN for layer-2. Report includes the address, desination, VNI, index, type, and last updated. |

## multicast-destinations

|             |                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>bridge-table multicast-destinations</b> |
| Tree        | <b>multicast-destinations</b>                                                                                                    |
| Description | Show a multicast-destinations vxlan interface report for EVPN-VXLAN for layer-2. Use with the destination or VNI options.        |

## destination | VNI

|             |                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>bridge-table multicast-destinations destination   VNI</b>                                    |
| Tree        | <b>destination   VNI</b>                                                                                                                                                              |
| Description | Show a unicast-destinations vxlan interface report for EVPN-VXLAN for layer-2 for a specific destination or VNI or use the asterisk after either option to see all destinations/VNIs. |

## unicast-destinations

|             |                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>bridge-table unicast-destinations</b> |
| Tree        | <b>unicast-destinations</b>                                                                                                    |
| Description | Show a unicast-destinations vxlan interface report for EVPN-VXLAN for layer-2. Use with the destination or VNI options.        |

---

## destination | VNI

|             |                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>bridge-table</b><br><b>unicast-destinations destination   VNI</b>                            |
| Tree        | <b>destination</b>                                                                                                                                                                    |
| Description | Show a unicast-destinations vxlan interface report for EVPN-VXLAN for layer-2 for a specific destination or VNI or use the asterisk after either option to see all destinations/VNIs. |

## brief

|             |                                                                                                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>brief</b>                                                                                         |
| Tree        | <b>brief</b>                                                                                                                                                                               |
| Description | Show a brief vxlan interface report for EVPN-VXLAN for layer-2. The brief report includes the tunnel-interface, VXLAN-interface, type (bridged/routed), ingress VNI, and egress source-ip. |

## detail

|             |                                                                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Context     | <b>show tunnel-interface</b> <i>interface</i> <b>vxlan-interface</b> <i>interface</i> <b>detail</b>                                                                                                                                                                        |
| Tree        | <b>detail</b>                                                                                                                                                                                                                                                              |
| Description | Show a detailed vxlan interface report for EVPN-VXLAN for layer-2. The detailed report includes the same details as the brief report (tunnel-interface, VXLAN-interface, type (bridged/routed), ingress VNI, and egress source-ip), plus bridge table and summary details. |

---

# 7.11 Version show reports

[show](#)  
— [version](#)

## 7.11.1 Version descriptions

### show

|             |                             |
|-------------|-----------------------------|
| Context     | <a href="#">show</a>        |
| Tree        | <a href="#">show</a>        |
| Description | Container for show reports. |

### version

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| Context     | <a href="#">show version</a>                                                                                 |
| Tree        | <a href="#">version</a>                                                                                      |
| Description | Show a version report that contains the currently running software version, last booted, and memory details. |

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